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THE ACOUSTIC IMPENDANCE OF PERFORATES AT MEDIUM AND HIGH SOUND PRESSURE LEVELS

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A study has been carried out of the behaviour of the acoustic impedance of a range of perforates at medium and high incident sound pressure levels. It included a detailed experimental programme together with a theoretical analysis of the problem. The theoretical work yields, for the linear impedance range, an improved formula for the impedance of perforates which is based on that for a single isolated orifice modified for interaction effects. This prediction technique is substantiated by the results of the experimental programme. Theoretical modelling of the non-linear impedance is based on a quasi-steady approximation to the acoustic flow through the orifice; this approach yields a formula for the non-linear acoustic resistance based on steady flow data for the perforate, which is readily obtainable. Further, the analysis yields a method of predicting when the non-linear mass reactance attains its asymptotic minimum. The correlation between experimental and theoretical results is good.

Other associated aspects of the work are discussed in some detail. These include (i) the coupling of the mechanical and acoustic impedance of the perforated plate, (ii) the effects of honeycomb in the backing cavity and (iii) the two-microphone method of impedance measurement. Finally, some consideration is given both to modifying the non-linear behaviour of perforates and to applications of the prediction formula where the pressure rather than the velocity field is specified.

1. INTRODUCTION AND FORMULATION OF THE PROBLEM

1.1. INTRODUCTION

In recent years a tremendous amount of manpower and effort has been spent on the study of the behaviour of acoustically absorbing materials. In particular, the emphasis has been on materials suitable for aero-engine applications. The study in relation to aero-engines has, of course, embraced all aspects of the acoustic treatment of ducts, including the behaviour of the material and the effects of geometry and air flow. That particular aspect of the material behaviour which is the primary concern in the work contained herein, is the sensitivity of the acoustic impedance of the material to incident sound pressure level. The sound pressure levels at which this sensitivity occurs depend, in general, on the type of material. For many materials met in practice these changes in impedance become apparent above about 130 dB (*re* 0·0002 dyne/cm^2). The changes in impedance usually comprise an increase in the real part of the impedance of the material, together with a decrease in the imaginary part.

The non-linear behaviour of many materials is quite marked, especially if they are of the perforated plate type. The non-linear part of the impedance, in general, dominates the acoustic behaviour of the material at the sound pressure levels of interest in aero-engine

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applications. Thus an ability to predict and understand this non-linear behaviour is essential if accurate predictions of duct attenuation rates for aircraft engine liners are to be made. These duct attenuation rates are generally sensitive both to lining material parameters and to the type of noise source. Therefore, they are important factors in predictions of noise levels to which the population is exposed. Liner design is perhaps even more important to the engine designer as a factor affecting engine weight and performance. Further, it has been established experimentally that a liner performance is affected by air flow over its surface and it is possible that these flow effects and the non-linear sound pressure level effects are interdependent. Therefore, clarification of the nature of the dependence on sound pressure level will assist in identifying and understanding these flow effects.

1.2. SOME BASIC ASPECTS OF SOUND ATTENUATION IN LINED DUCTS

To establish a basis for formulating the whole problem quantitatively, specific consideration must first be given to certain, reasonably well-known, fundamental aspects of sound propagation in attenuating ducts carrying flow.

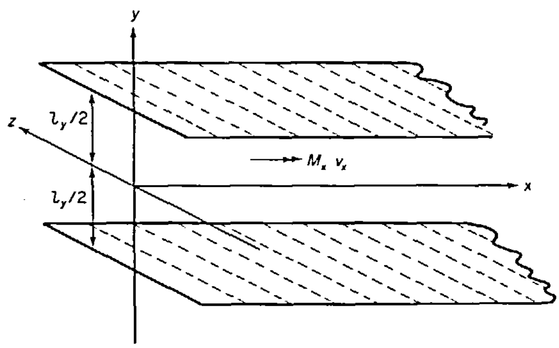

Figure 1. Coordinate system for attenuating duct model with flow. $M_x = v_x/c$, where v_x is the mean flow velocity in the x direction and c is the speed of sound.

For an infinite, two-dimensional duct (the three-dimensional case does not involve any essentially different behaviour), carrying a sheared mean flow of an inviscid fluid, the general equation governing the transverse amplitude distribution function, $F(y)$, of a propagating acoustic pressure disturbance is (see reference [1])

$$\frac{d^2F(y)}{dy^2}+2Y\frac{dM_x}{dy}\frac{\partial F(y)}{\partial y}+[(k-k_xM_x)^2-k_x^2]F(y)=0, \qquad (1)$$

where the notation in the cited equation has been modified by writing $k+j\alpha=k_x$ and $Y=k_x/(k-k_xM_x)$. k is the free space wave-number, k_x is the wave-number in the x-direction and M_x (a function of y for the sheared flow) is the Mach number of the mean flow in the x-direction. The pressure and the transverse particle velocity (in the y-direction, that is—see Figure 1) are, respectively,

$$p=F(y)\exp[j(t-k_xx)],$$
$$v=G(y)\exp[j(t-k_xx)]. \qquad (2)$$

The particle velocity amplitude distribution function, $G(y)$, can be related to $F(y)$ by using the linear momentum equation for the fluid (see reference [1]). This relationship is

$$G(y)=-\frac{1}{j\omega\rho}\left(1-\frac{k_xM_x}{k}\right)\frac{\partial}{\partial y}[F(y)]. \qquad (3)$$

For simplicity, only disturbances propagating in the positive x-direction are considered. For the illustrative purpose of the discussion here it is sufficient to consider only cases where the mean shear term in equation (1) (the second term, proportional to dM_x/dy) is important only in a relatively thin layer at the wall: i.e., when $M_x(y)$ can be taken, in the limit, as constant across the duct and dropping abruptly to zero at the wall. In aircraft engine ducts the boundary layer often is not thick compared with relevant acoustic wavelengths, and recent work by Vaidya [2], Mungur [3], Tester [4], Dean [5] and others has indicated that reasonably accurate results can be obtained by using a uniform "plug" flow model of this limiting type with an appropriately modified wall impedance, the modification expressing the effects of the shear flow on the no-flow acoustic impedance. Thus, a general solution for $F(y)$, when $M_x(y)$ has a "plug" flow behaviour, is

$$F(y) = A\sin(k_y y) + B\cos(k_y y), \tag{4}$$

where the transverse wave-number, k_y, is defined by $k_y^2 = (k - k_x M_x)^2 - k_x^2$.

The transverse wave-number, k_y, is determined by the boundary conditions: namely, that the ratio of the acoustic pressure to the normal acoustic particle velocity (into the wall) at the walls, $y = \pm l_y/2$, must equal the specific normal acoustic impedance. This normal impedance in the general case is a function of the angle of incidence since, in general, waves may propagate in the liner material both normal to and along the axis of the duct. In the specific case of interest here the acoustic particle velocity in the liner material is effectively constrained to be normal to the face of the liner, as a result of the honeycomb backing to the liner. This assumption may also be carried over to the non-linear impedance situation, for single frequency excitation. The single frequency non-linear impedance then remains meaningful in many situations of multi-frequency excitation of practical interest, in which interactions between different spectral components of the signal are relatively weak.

In view of the preceding discussion, a simple impedance boundary condition can be assumed here: that is, at the walls ($y = \pm l_y/2$),

$$\frac{p}{v} = Z\rho c = \frac{F(y)}{G(y)}\,\frac{k - k_x M_x}{k},$$

where Z is the specific normal acoustic impedance ratio, which is presumed to be known in terms of the geometry and composition of the liner material, and appropriate mean flow parameters. Substitution for $\partial[F(y)]/\partial y$, from equation (3), and simplification, by using equation (4), gives

$$j\frac{kl_y}{2}\left[1 - \frac{k_x M_x}{k}\right]^2 \Big/ Z = -\frac{k_y l_y}{2}\,\frac{\cot}{\tan}\left(\frac{k_y l_y}{2}\right),$$

where the cot solution gives the even modes and the tan solution the odd modes. The equation becomes simpler, but retains its essential character, in the case of zero mean flow, reducing to

$$j\frac{kl_y}{2}\frac{1}{Z} = -\frac{k_y l_y}{2}\,\frac{\cot}{\tan}\left(\frac{k_y l_y}{2}\right), \tag{5}$$

with the transverse wave-number now given by

$$k_y^2 = k^2 - k_x^2. \tag{6}$$

Solving equation (5) yields k_y and substitution of this result into equation (6) yields k_x. From equation (2) it is evident that the pressure attenuation per unit length depends upon the imaginary part of k_x. From equation (6)

$$k_x l_y/2 = \pm\left[\left(\frac{kl_y}{2}\right)^2 - \left(\frac{k_y l_y}{2}\right)^2\right]^{1/2}. \tag{7}$$

Thus a convenient non-dimensional expression for the attenuation is in the form of dB/duct-width (or per $\frac{1}{2}$ duct-width, as in this case with equal impedances on the opposite walls). The attenuation, in its non-dimensional form, can be seen from equations (7) and (5) to be controlled by the wall impedance.

It has been shown, theoretically, by Cremer [6] that an optimum wall impedance exists which will produce a maximum attenuation per unit length (or "attenuation rate") in the duct for the least damped mode: viz., $Z_{opt} \simeq \rho c(0{\cdot}93 - j0{\cdot}74)(l_y/\lambda)$. This expression, together with equations (5) and (7), shows that this optimum attenuation rate is dependent on both real and imaginary parts of the impedance. Parametric studies (see reference [7]) have indicated that, in general, nominal changes in the reactive part of the impedance from the optimum value will result, primarily, in a shift of the optimum attenuation frequency—with, of course, a change in attenuation from the optimum value—whilst a change in resistance from the optimum value will result, primarily, in lower attenuation rates. Generally, if the resistance changes to a value greater than the optimum value an increase in attenuation bandwidth will result. Calculations for practical, perforated-plate-type liners often exhibit relatively narrow bandwidth attenuation. Thus, in general, the wall impedance needs to be carefully chosen to maintain high attenuation over a given band.

Similar considerations may also, in general, be carried over to the case of non-linear impedance, provided, of course (as mentioned above), that the impedance is defined for a specific frequency, and that the propagation within the duct can be written in terms of the usual linear equations. As a result of both the non-linear behaviour of the material and the general duct attenuation properties outlined in the analysis above, three especially significant trends become evident. These are as follows.

(a) The real part of the non-linear impedance is, in general, an order of magnitude higher than the nominal, linear value of the resistance of the material for the sound pressure levels typical of in-engine applications.

(b) Because the wave is being attenuated as it propagates, the wall impedance is changing (since the impedance is dependent on the sound pressure level); thus, the attenuation rate is changing (see equations (5) and (6)).

(c) If the liner impedance were chosen to be an optimum at a certain frequency, then, due to attenuation of the pressure wave, the wall impedance would change with consequential loss in attenuation rate relative to the optimum.

The discussion thus reveals the specific reasons for accurate prediction of the impedance, both for predicting the liner attenuation rates anticipated in practice and also for designing for high attenuation rates at certain frequencies.

It should also be noted that the solution here for the attenuation rates has been outlined only for the infinitely long duct case. In practice, of course, any test section will have only a finite length and therefore have terminating conditions different to those assumed here, because of the presence of reflected waves at the termination. For measurement stations external to the duct (and indeed within the duct) the measured attenuation may therefore be somewhat different from that calculated on the assumption of a non-reflecting duct termination.

This may be stated more specifically: the insertion loss will only equal the transmission loss when the duct is terminated in its characteristic impedance. Here, the "transmission loss" is defined as the calculated value for reflection-free termination conditions and the "insertion loss" as that measured for any particular practical termination condition. Whilst, in principle, the insertion loss may be calculated if the termination conditions are known, this is not usually a satisfactory procedure in practice, due to the presence of other factors, including the high densities of excited duct modes generally met with in the practical situations.

It is therefore necessary to review practical liner test facilities, to examine the influence of a range of practical terminating conditions on the measured performance of the liner, and, if possible, to devise a method of correcting for this influence. Such an investigation has been carried out and reported [8, 9]. Two principal features of the analysis are that it includes the following effects: (i) that of flow on the acoustic fields as well as on the liner absorption; (ii) that of possible variations in the impedance of the terminating chamber.

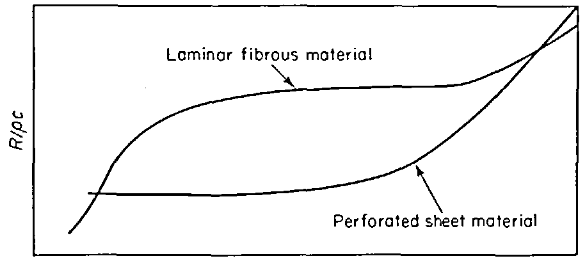

Figure 2. Illustrative comparison of the behaviour of the specific acoustic resistance ratio of a perforate and fibrous laminar material (ρc termination conditions).

Finally, the non-linear impedance of a liner material depends on the parameters characterizing the type of material. There are basically three types of material which may be used for liner applications: perforated plate, bulk absorber, or laminar bulk material (e.g., "Feltmetal"). To the aero-engine manufacturer the perforate is most attractive because of its low cost, self cleaning/draining ability and ease of fabrication, although it does tend to have the "worst" non-linear behaviour (see, for example, Figure 2). Initially it was hoped to carry out a study of the non-linear behaviour of both perforate-type absorbers and the laminar bulk types, and to determine the governing parameters for both types. However, severe delays in the manufacturing of the impedance facility made it necessary to concentrate on one particular type. The choise of material for study was the perforated plate type, because of its preferential use to date.

The perforated plate, as distinct from a collection of isolated holes, has received relatively little attention, either theoretically or experimentally, it generally being assumed that by correcting the single hole values to take account of porosity a sufficiently accurate result would be obtained. The work carried out on single holes, whilst in itself by no means complete, at least suggests some of the parameters which may be relevant in describing the behaviour of perforates: for example, Reynolds number, or acoustic Mach number (i.e., acoustic particle velocity) and perhaps Strouhal number based on hole diameter.

1.3. IDENTIFICATION AND FORMULATION OF THE RESEARCH PROGRAMME

Several previous investigations of the non-linear behaviour of single isolated holes have been reported (see, e.g., references [10–13]). These, while not fully satisfactory, provide a fair amount of information on the nature of the dependence of the plate/hole system impedance on the relevant parameters.

However, the behaviour of perforates with closely spaced holes might be anticipated to have some characteristics that are quite distinct from those of a model based on a set of isolated non-interacting holes, both in the linear and non-linear regimes. Simple physical arguments suggest that interaction could cause a reduction in both the attached mass and the resistance (see section 2.5 and Figure 5). None of these possible effects have been previously investigated or suggested in the literature. For example, Zwikker and Kosten (see reference [14]) state that the impedance of an array of holes is equal to that of a single isolated hole divided by the porosity. No mention of possible interaction effects, which may be significant especially at the higher porosities, is given. Thus it is of interest to examine this aspect of behaviour both in the linear and non-linear regions of behaviour, but particularly in the latter case.

Further, the previous studies were ideal, in that the hole, whilst having a small thickness compared with diameter, was machined from a thick plate, the plate being chamfered at the hole to obtain the desired thickness. Because of its thickness, the plate was effectively rigid, and hence its motion could be disregarded. In the cases of interest here (i.e., thin perforated plates of porosities not less than $7\frac{1}{2}\%$) plate motion is not necessarily negligible. Indeed, with small hole sizes, the manufacturing process imposes a limitation on the maximum plate thickness. In general, the plate thickness may not be greater than the hole diameter.

Other fabrication limitations arise for samples with high porosities and small hole sizes, For example, some 2000 holes may be required in a 2 in $\times$ 2 in sample. The chamfering technique is not practical owing to the small hole diameter/hole spacing ratio (1:2) in high porosity samples. The selection of samples thus must be restricted to those available from manufacturers.

The non-linear behaviour is, of course, of major interest here as this, in general, controls the impedance of the sample in an aircraft engine duct. It is particularly interesting that, even in the case of a single isolated hole, no completely satisfactory empirical or theoretical model has been presented for the behaviour. Thus, the main objective of the work is to clarify both interaction and non-linear effects, with emphasis on obtaining, ultimately, engineering design information. The work must therefore be closely tied to practical situations, the sample specifications being typically those which might be used in an in-engine application.

In the light of these considerations the following programme objectives were determined.

The first objective was to effect the design and construction of a high-precision impedance and attenuation (insertion-loss) test facility, to specifications based on existing knowledge of those parameters relevant to the study of the non-linear behaviour of liner materials. (This test facility is described in reference [15].)

The second objective was to carry out normal impedance measurements on a range of perforates over a specified range of incident sound pressure levels. The samples chosen were typically those which might be used in in-engine applications, having hole diameters 0·026 in, 0·050 in and 0·110 in, respectively. The porosities chosen were $7\frac{1}{2}\%$, 15% and 22%; a common plate thickness of 0·022 in was selected. Measurements were to be carried out at three test frequencies consistent with quarter wavelength cavity depths of 1 in, 2 in and 3 in. In practice, the maximum sound pressure levels available for testing would be limited by the maximum usable transducer driver coil current† (beyond this current setting, experience showed that the coils tended to break up rather quickly).

The third objective was to carry out a theoretical analysis of the expected non-linear behaviour of the impedance, this analysis having the aim of providing a method of prediction of the non-linear behaviour from readily available (or at least easily acquired) data on the material: e.g., prediction based, say, on parameters such as steady flow discharge coefficient, hole spacing and the like.

The fourth objective was to examine and analyse the design and performance of practical test facilities for the evaluation of liner performance, and the methods of extraction of data from these. (This study has been reported in reference [9].)

The overall objectives of the work programme, taking all these various aspects together, were (i) to provide a practical method of prediction of liner behaviour requiring only readily

† The only available, suitable transducers were Ling Altec 290E units (see reference [15]).

available engineering data, and (ii) to provide physical insight into the mechanism of non-linear behaviour *via* a detailed, controlled, precision measurement programme.

Two further pieces of work, which were not specifically associated with the non-linear behaviour of perforates but which arose within the general context of the programme, were also carried out. The first of these was a design feasibility study for a portable liner test device (which could be easily operated by normal ground crews) to monitor a liner's in-engine condition. By necessity it had to be a simple self-contained device. This device is described in reference [16] (see Appendix I). In the second related study the so-called "two-microphone" method of impedance measurements was investigated. Considerable time was spent investigating the accuracy and limitations of this technique. The impedance tube provided an excellent test set-up as the impedance could be determined by both techniques simultaneously. The technique was extremely attractive both because of the reduced time required to test a sample, and because of the simplicity of the measuring procedure. Also, if it proved satisfactory, it could provide a compact technique for measuring liner impedance *in situ* with air flow present, a measurement which is difficult to make and which urgently requires suitable techniques. Discussion of some aspects of this work is included in this paper; the basic principles of the method are outlined in reference [15].

A separate historical review of previous experimental and theoretical work is not included. Relevant references to, and thorough discussions of, previous work are given in each section, as appropriate. In particular, previous work on the normal impedance of perforated plates for the linear regime is critically evaluated in section 2, and used to establish a generally valid, accurate formula for this impedance.

2. THE NORMAL IMPEDANCE OF PERFORATED PLATES: LINEAR REGIME

2.1. INTRODUCTION

Acoustic absorption in materials is fairly well understood (see, for example, reference [14]) for situations in which the sound pressure levels are not too high (generally, less than 100 dB). The practical performance of a material as an absorber is usually quantified in terms of its ability to absorb randomly incident energy. The random incidence absorption coefficient can generally be developed from the normal incidence absorption coefficient of a material [17], which is defined as the ratio of the energy absorbed by a material to the energy incident upon it, for a normally incident plane wave. This normal absorption coefficient itself, however, is not a fundamental parameter of the material, in that it, in turn, may be derived from the normal acoustic impedance of the material. The normal acoustic impedance is the complex ratio of the acoustic pressure to the normal acoustic particle velocity at the surface of the material, the normal being in the direction into the material. In many cases this normal acoustic impedance can be completely specified in terms of physical and geometrical parameters of the material: for example, for bulk absorbers, the complex propagation constant and characteristic impedance of the material, which, of course, are in turn dependent on both the geometry and the physical properties of the basic, actual materials making up the acoustic absorbing "material". It may similarly be defined for perforated plates (as will become apparent) in terms of their geometry, and material parameters of the medium, such as speed of sound, and viscosity and thermal conduction coefficients. Thus, for the general case it is possible—provided certain material parameters are available—to define the normal incidence acoustic impedance and hence deduce the appropriate absorption coefficient (this is true irrespective of whether the material is bulk, Helmholtz or composite in type).

The normal incidence absorption coefficient has its main usefulness in the practical screening of materials for architectural applications. In determining the desirability of a material as a duct liner the impedance of the material (as indicated in the theory of section 1) is the more relevant material parameter. (Although, for example, the Sabine formulation for duct attenuation depends on the random incidence absorption coefficient, this formulation, of course, is a limiting case of the general theory referred to above.)

Thus the emphasis of this work is on determining the behaviour of the normal impedance of the material, both as a function of frequency and, more particularly, of sound pressure level.

The non-linear behaviour of the impedance of a single hole in a plate has been known for some considerable time (see, for example, reference [10]). As a perforated plate is simply a collection of single holes, its impedance would also be expected to exhibit non-linear behaviour, and in addition, interactions amongst the holes might affect this behaviour. In order to optimize, or to even compute accurately, the attenuation provided by a particular liner, it is essential to have a knowledge of the non-linear behaviour of the liner material, particularly if the non-linear behaviour is strongly dependent on sound pressure level and is therefore controlling the impedance of the material.

Because of the lengthy process of experimentally determining the normal impedance of materials—particularly when these depend upon sound pressure level—the engineer requires a method of rapidly assessing materials in order to rank them for design purposes. Ideally, he requires a technique of predicting their non-linear behaviour based on either readily available or, alternatively, easily obtainable, data such as discharge coefficients or steady flow resistance values. Such techniques would appear to be practical for the perforated plate type absorbers under consideration here. In the following section the development of a parametric expression for the normal acoustic impedance of a perforated plate will be carried out, based on the theory developed for single holes. In this initial development, only the linear regime of behaviour will be considered (i.e., where the impedance is independent of sound pressure level); this is necessary as the linear models presented in the available literature to date are either inadequate or incorrect. Subsequently, in section 4, the models for the non-linear behaviour will be developed. Of particular interest throughout are those models based on steady-state data, and also the behaviour of the resistive part of the impedance, this being most strongly influenced by sound pressure level and in general governing the attenuation behaviour of the liner.

The impedance of a regularly perforated plate can be quantified, for the linear regime, in terms of the non-dimensional hole diameter, kd, the non-dimensional plate thickness, kl (where k is $2\pi/\lambda$, the wave-number), an appropriate dissipative parameter, F (involving fluid viscosity and/or thermal conductivity coefficients), and, for holes of given surface density, the number of perforations per unit area. Alternatively, the porosity (P) may be used in preference to the number of perforations per unit area. (The porosity is defined as the ratio of open to total area.) Thus $Z_{perf} = \phi(kd, kl, P, F)$, evaluated at some frequency, say, $f = c/\lambda = kc/2\pi$.

Both the resistive and reactive parts of the impedance may be frequency dependent. In particular, the frequency dependence of the resistive part of the impedance, composed of both viscous and radiation impedance components (the latter dependent on hole/baffle geometry) will be subject to joint limiting conditions on frequency and hole geometry, as will become apparent. Similar comments also apply to the total inertial reactance which is composed of both mass and viscous inertial components.

It is convenient to consider, initially, the case of a single hole in a plate (of infinite extent) and to develop from this the concept of a total inertial reactance.

Lord Rayleigh [18] considered the inertial reactance of simple holes. In his analysis he considers the problem in terms of the velocity potential of the field around the hole. Because of the similarity with the electrical case Rayleigh frequently talks in terms of "resistance".

This resistance is not related to any viscous losses; on the contrary, it is related to what is now termed the mass reactance. This analogy of resistance introduced by Rayleigh leads naturally to the concept of conductance. Rather than use Rayleigh's notation for the development of the conductance analogy his method will be applied and an independent modern set of symbols and nomenclature will be used.

Consider the case of a single hole in a plate (see Figure 3). Let the area of the hole be denoted by S ($= \pi d^2/4$). If the plate thickness is small compared with the wavelength (and the hole diameter), then, under the influence of the incident wave, the air in the hole will move *en masse*; i.e., it will behave as a small piston of air. The mass of air in the hole is $\rho S l'$, where ρ is the density and l' is some effective hole length not yet defined.

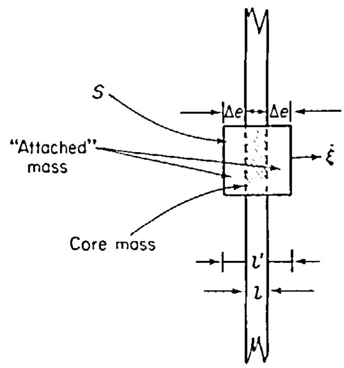

Figure 3. Schematic drawing of actual and attached mass of air in the hole, for a single hole in an infinite plate looking into free space. $S = \pi d^2/4$; $l' = l + 2\Delta e$ = effective length; Δe = end correction due to attached mass; l = plate thickness.

The kinetic energy (T) of this piston of air is $\rho S l' \dot{\xi}^2/2$, where $\dot{\xi}$ is the velocity. This may, in turn, be written as a function of the volume velocity through the hole, $S\dot{\xi}$, whence,

$$T = \tfrac{1}{2}\frac{\rho l'}{S}(S\dot{\xi})^2,$$

or,

$$T = \tfrac{1}{2}\rho\frac{1}{K}(S\dot{\xi})^2.$$

K, the conductance of the hole, is a function of the parameters of the hole. The term "conductance" arises from the similarity of the energy term with electrical circuits: if one takes the term in front of the volume velocity to be analogous to resistance, then the reciprocal of this is the conductance. So far in the treatment of the problem, reference has been made to an effective length for the mass of air in the hole. This effective length is composed of three components: the "core-mass", $\rho S l$, and two end-corrections, each of an amount $\rho S \Delta e$. This effective end-correction, Δe, shown by Rayleigh for the case under consideration to be $4d/3\pi$, is a result of coupling—through the local pressure field—between the vibrating core-mass and the air external to the hole: the vibrating core-mass, in effect, pushes this additional mass. The important point to note is that it is an *end-effect*. The so-called core-mass is an obvious component of the total effective mass, evident from an elementary treatment of the dynamic behaviour of air in a tube or hole. The end-correction is not.

2.2. CRANDALL'S THEORY OF THE IMPEDANCE OF A SINGLE TUBE

The next step is to carry out a general detailed treatment of the air confined in the hole or aperture, including viscous and thermal effects. A modern general treatment was first carried out by Crandall [19]; Stokes, Helmholtz, Kirchoff and Rayleigh had previously solved the problem [18, see Volume II, Chapter XIX].

Crandall's treatment, which was derived for the linear regime of behaviour, yields directly two limiting cases for resistive losses: namely, Poiseuille-type and Helmholtz-type, respectively (the latter being frequency dependent). The theoretical model considers only the fluid which is contained within the aperture or tube, and is therefore exclusive of any end-effects.

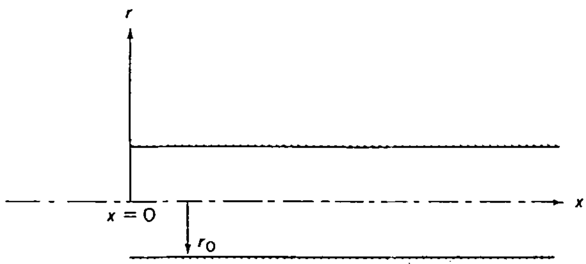

Figure 4. Coordinate system for Crandall's theoretical model for viscous effects in tubes.

Crandall considered a tube of fluid (Figure 4) in which a shear layer is set up as a result of viscous retardation of fluid near the wall. If the axis of the tube be denoted as x, and the pressure gradient parallel to the axis of the tube as ϕ, then the total driving force on an annular ring of fluid, at radius r, can be written as $(\phi\,\mathrm{d}x)2\pi r\,\mathrm{d}r$. This is balanced by the inertial and resistive forces which are, respectively, $j\omega\rho 2\pi r\,\mathrm{d}r\,\mathrm{d}x$ and $\partial/\partial r(-2\pi r\,\mathrm{d}x\,\mu\,\partial\dot{\xi}/\partial r)\mathrm{d}r$, where μ is an effective coefficient of viscosity (in which thermal conductivity effects can be included) and $\dot{\xi}$ is the particle velocity (the negative sign is due to the negative pressure gradient).

The equation of motion can be written down immediately, as

$$\left[j\omega\rho - \frac{\mu}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right)\right]\dot{\xi} = \phi.$$

As $\dot{\xi}$ is a function of r only, this equation may be rewritten as

$$\left[\frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + k_s^2\right]\xi = -\frac{\phi}{\mu},$$

where $k_s^2 = -j\rho\omega/\mu$, the solution of this being

$$\dot{\xi}(r) = -\frac{\phi}{\mu k_s^2} + AJ_0(k_s r),$$

where J_0 is the zero order Bessel function. From the boundary condition that the velocity must vanish at the boundary, $r = r_0$, the constant, A, may be determined. Finally, the velocity $\dot{\xi}(r)$, may be integrated over the area of cross-section to give the mean velocity:

$$\dot{\xi} = \frac{2}{r_0^2}\int_0^{r_0}\dot{\xi}r\,\mathrm{d}r = -\frac{\phi}{\mu k_s^2}\left[1 - \frac{2J_1(k_s r_0)}{k_s r_0 J_0(k_s r_0)}\right].$$

The quantity in the square brackets is thus a velocity profile function, where J_0 and J_1 are Bessel functions of order zero and one, respectively. The pressure difference across the tube ends is

$$p = \int_0^l \phi \, dx.$$

Using this expression together with that for the mean velocity gives, for the input impedance per *unit* cross-sectional area of the tube,

$$z' = -\mu k_s^2 l \Bigg/ \left[1 - \frac{2J_1(k_s r_0)}{k_s r_0 J_0(k_s r_0)}\right],$$

or

$$z' = j\omega\rho l \Bigg/ \left[1 - \frac{2J_1(k_s r_0)}{k_s r_0 J_0(k_s r_0)}\right], \tag{8}$$

since $k_s^2 = -j\omega\rho/\mu$. k_s is the wave-number of the viscous Stokes wave and λ_s is the associated wavelength, which can be written, where $\beta = (\rho\omega/2\mu)^{1/2}$, as $\lambda_s = 2\pi/\beta$. In air, for a thermally insulated wall, $\mu = 2 \times 10^{-4}$ poise, so that, at 1000 Hz, $\lambda_s \simeq 0{\cdot}04$ cm or 0·016 in. For a wall of high thermal conductivity, the effective value of the viscosity coefficient is $\mu = 4 \times 10^{-4}$; thus the Stokes wavelength in this case is greater by a factor of $\sqrt{2}$.

Depending on the value of the Stokes number, $|k_s r_0|$, the above equations (8) have two limiting values.

(a) If $|k_s r_0| < 2$ (i.e., when $r_0 < \lambda_s/\pi$, which is satisfied for air if $r_0^2 f < 1{\cdot}0$, r_0 being in cm), then a series expansion for the Bessel functions may be used. Taking the first two terms only gives

$$z' = \frac{8\mu l}{r_0^2} + j\frac{4}{3}\rho\omega l = R + j\chi. \tag{9}$$

The expression $8\mu l/r_0^2$ is immediately recognizable as Poiseuille's law of resistance for laminar flow of viscous fluids in narrow tubes. The reactive part of the impedance, $\chi = 4\rho l\omega/3$, has a total effective mass $4\rho l/3$, which is $\frac{1}{3}$ greater than the actual mass in the tube per unit area. Although explicitly independent of the viscosity coefficient, this additional attached mass is a direct consequence of the influence of viscosity on the velocity profile.

(b) If $|k_s r_0| > 10$ (i.e., for $r_0^2 f > 5$ in air), then the appropriate approximation to the Bessel function ratio is $J_1(x\sqrt{-j})/J_0(x\sqrt{-j}) = -j$. Consequently, it may be shown that

$$z' = \frac{l}{r_0}\sqrt{2\rho\mu\omega} + j\omega\rho l\left[1 + \frac{1}{r_0}\sqrt{\left(\frac{2\mu}{\rho\omega}\right)}\right]. \tag{10}$$

The real part of the impedance is now frequency dependent. This expression for the resistance was first determined by Helmholtz. It is also noteworthy that the additional "attached mass" —a result of the combined viscous and inertia effects in the tube—is also frequency dependent.

It must be borne in mind that the formulae presented in equations (9) and (10) above are for unit area of the hole: i.e., they represent the specific acoustic impedance of the hole. If, for a collection of holes, there is no interaction between the holes and if each hole, of area σ, has an associated zone area S, then, from continuity of pressure and volume velocity at the interfaces, it can be shown that the specific acoustic impedance, z, referred to the hole array, is given by

$$z = z'\frac{S}{\sigma} = \frac{z'}{P},$$

where P is the porosity. (It has, of course, been assumed here that the array of holes is symmetrical and that there is no attached mass due to end-effect.)

2.3. END-CORRECTIONS: RESISTANCE

The formulation given in equations (9) and (10) is strictly applicable only to the case of the infinitely long tube. For the case of tubes of finite length (and so for perforated plates) an additional end-correction term must be added. The need for an end-correction term applies equally to the resistive and the reactive components. It is convenient, however, to examine the end-corrections for the resistive and reactive terms independently. These will be examined in the light of the two most significant contributions to the understanding of the problem, those of Sivian [10] and Ingard [20]. These two studies apply to the case of the single aperture.

Sivian [10] carried out measurements on a series of orifices—not too dissimilar in size from those considered here—at various sound pressure levels and frequencies. In order to allow for end-effects, Sivian wrote the total aperture resistance R as the sum of internal and external end-effect resistances, respectively: i.e., $R = R_i + R_e$, where R_i was the component due to viscous losses in the hole and R_e was due to viscous losses external to the hole. Further, it was assumed that the external part of the resistance had the same parametric dependence as that of the internal part, with the exception of two factors: viz., the viscosity is replaced by a value appropriate to a non-thermally conducting medium and the orifice length is replaced by some effective length.

Thus, since from equation (8),

$$z' = l[j\omega\rho]/\square(k_s r_0),$$

where

$$\square(k_s r_0) = \left[1 - \frac{2J_1(k_s r_0)}{k_s r_0 J_0(k_s r_0)}\right],$$

then

$$R = R_i + R_e = \mathrm{Re}\{l_i[j\omega\rho]/\square(k_s r_0) + l_e[j\omega\rho]/\square(k_s' r_0)\},$$

where the hole length l_i refers to losses in the hole and the hole length l_e refers to losses external to the hole. Similarly, k_s and k_s' refer to the viscous Stokes wave-number in the hole and that in the region external to the hole, respectively (i.e., μ for the conducting wall and μ' for the non-conducting wall, respectively).

In choosing a value for l_e, Sivian made the assumption that the viscous resistance end-correction was the same as the mass end-correction. He used the value derived by Rayleigh: viz., $16r_0/3\pi$, representing an additional effective attached length at *each* end of the orifice of approximately $0{\cdot}85r_0$. Thus the general expression for the nominal resistance of an orifice becomes

$$R_{\mathrm{nom}} = \mathrm{Re}\{l_i[j\omega\rho]/\square(k_s r_0) + 16r_0[j\omega\rho]/3\pi\square(k_s' r_0)\}. \tag{11}$$

This expression may be simplified in the two limiting cases, by using equations (9) or (10), as appropriate. Sivian found that equation (11) predicted the nominal values to within 17% or better, for the range of samples he measured.

Ingard [20] re-examined the problem both theoretically and experimentally. Ingard's work refers specifically to holes where viscous losses are of the Helmholtz type (b). He considers the additional loss term R_e as due to frictional losses over the parallel plate face surfaces surrounding the hole: i.e., $\frac{1}{2}\int R_s|u_r|^2\,\mathrm{d}S$, where $R_s = \frac{1}{2}(2\mu\omega\rho)^{1/2}$, u_r being the tangential velocity over the plate surfaces and the integral being carried out over both plate face surfaces.

This leads to a *total* end-correction length of r_0, as compared with a value of $1{\cdot}7r_0$ derived *via* Sivian's assumptions. However, in practice, Ingard found the best experimental value to be nearer $2r_0$ [20]. Therefore, on the assumption that this result, although verified only for Helmholtz-type losses, can be carried over to the general case (this is reasonable since the length, l, is an independent variable both in the general form of the resistance and the approximate, limiting cases), a "nominal", end-corrected resistance, R_{nom}, can be defined as

$$R_{\text{nom}} = \text{Re}\{(j\omega\rho)(l+2r_0)/\square(k_s r_0)\}. \tag{12}$$

This expression can be taken to represent all the "classical", linear resistance of the hole(s).

It is important to note that Ingard maintains the same coefficient of viscosity (i.e., that applicable to the air in the hole) throughout the calculation, whereas Sivian uses two separate viscosities. For the limiting cases of Poiseuille- and Helmholtz-type losses and the specific case of $l = 2r_0$ (this latter equality is generally applicable to the tests to be described subsequently), the predictions of the two formulations can be compared, as follows.

$r_0^2 f > 0{\cdot}5$: *Helmholtz-type losses* (*b*)

According to Ingard,

$$R_{\text{nom}} \simeq (l+2r_0)\sqrt{\mu} = {}^{\text{I}}R_{\text{nom}}.$$

According to Sivian,

$$R_{\text{nom}} \simeq l\sqrt{\mu} + 1{\cdot}7r_0\sqrt{\mu'} = {}^{\text{S}}R_{\text{nom}} = [l + 1{\cdot}7r_0\sqrt{(\mu'/\mu)}]\sqrt{\mu}.$$

For the losses in the hole, $\mu \simeq 4\times10^{-4}$ CGS units (heat conducting walls). For the losses external to the hole $\mu' = 2\times10^{-4}$ CGS units (heat insulating conditions). Now,

$$\frac{{}^{\text{S}}R_{\text{nom}}}{{}^{\text{I}}R_{\text{nom}}} = \frac{[l+1{\cdot}7r_0\sqrt{(2/4)}]}{(l+2r_0)},$$

which is equal, for $l = 2r_0$, to $3{\cdot}2/4 = 80\%$.

Thus the Sivian prediction is only 80% of the nominal value predicted by using Ingard's empirical factor.

$r_0^2 f < 0{\cdot}1$: *Poiseuille-type losses* (*a*)

According to Ingard,

$$R_{\text{nom}} \simeq (l+2r_0)\,\mu = {}^{\text{I}}R_{\text{nom}}.$$

According to Sivian,

$$R_{\text{nom}} \simeq \left(l + 1{\cdot}7r_0\frac{\mu'}{\mu}\right)\mu = {}^{\text{S}}R_{\text{nom}}.$$

Thus, for the case of $l = 2r_0$, the Sivian prediction is 75% of the Ingard prediction.

Whilst Ingard claims only a 9% difference between experimental and "theoretical" results, and Sivian 17%, it should be noted that the experimental techniques were quite different; this may have led to some differences between the two (Sivian used a 2-microphone technique which he claims was subject to errors of 10–20%).

It is worth noting in passing that, in the cases of interest, the usual radiation resistance (for a hole in an infinite baffle looking into free space), and consequently its addition to R_{nom}, is negligible since the specific acoustic radiation resistance, for this case, is given by $\frac{1}{4}(kr_0)^2$.

2.4. END-CORRECTIONS: REACTANCE

In general, the reactance of a hole will have two contributions: that due to mass inertia effects and that contributed from viscous effects (see Figure 3 and equation (9)). Reference has already been made to the inertia effects, both in the introductory section and in discussing the resistive end-correction. It was shown that the inertial component of the mass reactance was given by $\rho\sigma(l+2\Delta e)$, where σ is the area of the aperture, l its length and Δe some end-correction pertaining to each end. It can be shown that for a vibrating piston in a baffle ($\lambda \gg 2\sqrt{\sigma/\pi}$) there will be a resultant attached mass of $(8r_0/3\pi)\sigma$ at each end (this result was derived by Rayleigh [18]). Thus, the resultant inertial reactance can be written as

$$M_i = \sigma\rho[l+2(8r_0/3\pi)]$$

and the specific acoustic reactance is given by

$$m_i = \rho(l+16r_0/3\pi).$$

It should be noted that the correction factor, $16r_0/3\pi$, is independent of the "internal" attached mass value determined in equations (9) and (10) and is an additional factor which would not result from the theoretical approach presented there. Thus the total mass reactance for the limiting cases may now be written as follows.

2.4.1. $r_0^2 f < 0{\cdot}1$: *Poiseuille-type losses*

The specific acoustic impedance for the hole/aperture is given by

$$j\chi = j\omega m = j\omega\rho[\tfrac{4}{3}l + 16r_0/3\pi]. \tag{13}$$

2.4.2. $r_0^2 f > 0{\cdot}5$: *Helmholtz-type losses*

The specific acoustic impedance for the hole/aperture is given by

$$j\chi = j\omega m = j\omega\rho\left\{l\left[1+\frac{1}{r_0}\sqrt{\left(\frac{2\mu}{\rho\omega}\right)}\right]+16r_0/3\pi\right\}. \tag{14}$$

Since the acoustic wave-number is related to frequency by $k=\omega/c$, equations (13) and (14) may be written as

$$j\chi = j\omega m = j\rho ck[\tfrac{4}{3}l + 16r_0/3\pi], \tag{15}$$

and

$$j\chi = j\omega m = j\rho ck\left\{l\left[1+\frac{1}{r_0}\sqrt{\left(\frac{2\mu}{\rho\omega}\right)}\right]+16r_0/3\pi\right\}, \tag{16}$$

respectively. The quantity $\sqrt{(2\mu/\rho\omega)}$ is the viscous boundary layer thickness.

Sivian [10] found, in practice, that the above equations, when the viscous contributions to the mass reactance were negligible, agreed with the experimental results to within 10%. Ingard [20] derived a general expression for the attached mass of an aperture opening into a bounded space. Both his theoretical and experimental values, he found, led to the Rayleigh correction ($8r_0/3\pi$) for the limiting case of a single aperture in an infinite wall. The deviation between experimental and theoretical results, he claimed, was some 2·6%. The general form of the Ingard function is complicated; however, for the size of aperture under consideration the expression reduces to the more simple expression of Rayleigh, which proves to be adequately accurate. Bies and Wilson [21] found the Rayleigh end-correction to be in good agreement with experimental results.

If there is no interaction between holes, then, for an array of holes, with the hole area σ and an associated zone area S per hole, the specific acoustic reactance referred to in the array is found by multiplying equations (15) and (16) by S/σ (i.e., the inverse of the porosity).

2.5. THE EFFECT OF INTERACTION BETWEEN THE HOLES IN AN ARRAY: THE PERFORATED PLATE

In the generalization of the single aperture formulae (quoted previously) to the case of the perforate, it has been assumed that there is no interaction between the apertures. This statement is certainly true for widely separated holes, and a very limited amount of investigation has been carried out to establish this [20, 22]. Ingard [20] considered the case of two apertures interacting, for a particular value of $\sqrt{\sigma/S}$ of 0·15 and found that the end-correction was very dependent on the separation. This can be shown by a very simple physical argument. Consider two apertures, each of area σ and set in a total area S. Then, if there is no interaction, the reactive part of the specific acoustic impedance ratio, referred to the area S, is $\chi_I \sim (1/P)2\delta\sqrt{\sigma}$, where δ is the total end-correction for a simple aperture (it being assumed that the plate is infinitely thin). Now assume that the two areas are coupled together into a total area of 2σ. Then the specific acoustic mass reactance ratio becomes $\sim(1/P)\delta\sqrt{2\sigma}$: i.e., when the holes are well separated the mass reactance is a factor of $\sqrt{2}$ greater than when they are superimposed.

The important aspects of the problem of interaction are illustrated diagrammatically in Figure 5. Here a comparison between a single isolated hole and two interacting holes is made, with reference both to resistance and reactance (cases (b) and (a) of the figure, respectively). The interacting mass case (a) shows the idealized total mass distribution with the actual mass distribution superimposed; the double hole configuration illustrates how, in the actual distributed attached mass situation, the two regions overlap giving rise to so-called

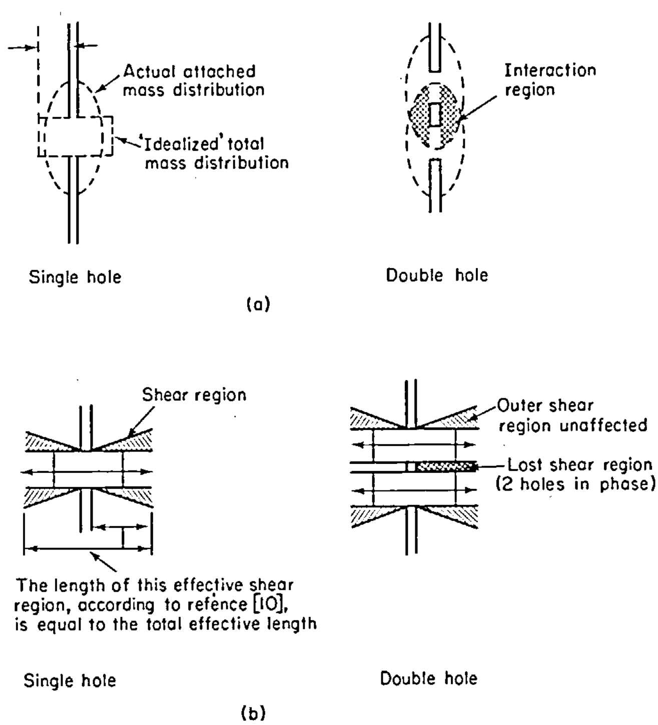

Figure 5. Idealized schemes for interaction effects, (a) for attached mass, showing single and two interacting holes and (b) for resistance end correction showing reduced shear region between single and two interacting holes.

interaction. For the resistive case the idealized mass distributions have been used to illustrate the effective shear region length as assumed by Sivian [10]. Clearly, in the two-hole situation the shear region is disturbed. It must be emphasized that these are mere convenient physical pictures of the situation. In practice, for the two-aperture problem [20], the interaction impedance is determined from

$$z_{1,2} = \frac{1}{u_2} \int_{S_2} P_{1,2}\, dS_2,$$

where $P_{1,2}$ is the pressure exerted at hole 2 by hole 1, u_2 is the volume velocity through hole 2 and S_2 its area. Thus, *via* this interaction impedance, the net impedance may be determined and a new end-correction factor determined.

The problem of acoustic interaction has been solved elegantly by Fok [22]. Fok carried out his solution for the case of an infinitely thin plate. He gives the attached conductance as $K' = 2r_0\psi'(\xi)$, where $\psi'(\xi)$ is now called the Fok function of the variable $2r_0/D$, where D is $2\sqrt{S/\pi}$, S being the zone area of each hole, and

$$\psi'(\xi) = 1 + a_1\xi + a_2\xi^2 + \cdots +,$$

where

$$\begin{aligned} a_1 &= -1{\cdot}4092, & a_2 &= 0, \\ a_3 &= +0{\cdot}33818, & a_4 &= 0, \\ a_5 &= +0{\cdot}06793, & a_6 &= -0{\cdot}02287, \\ a_7 &= +0{\cdot}03015, & a_8 &= -0{\cdot}01641. \end{aligned}$$

The function $\psi'(\xi)$ is shown in Figure 6. From Figure 6 it can be seen that if $2r_0/D < 0{\cdot}2$ there is no appreciable interaction effect and if $2r_0/D > 0{\cdot}8$, the attached mass is effectively zero.

It is now possible to modify the case of finite thickness plates for interaction effects, by modifying the end-correction with the Fok function, as follows. From equation (15), the specific acoustic reactance of the hole is

$$j\chi = j\omega m = j\rho ck(\tfrac{4}{3}l + 16r_0/3\pi),$$

and thus, for the Poiseuille dependent region,

$$j\omega m_I = j\rho ck\left[\tfrac{4}{3}\,l + \frac{16r_0}{3\pi}\bigg/\psi'(\xi)\right], \tag{17}$$

where m_I is the mass reactance modified for interaction effects. The reactive specific acoustic impedance, referred to the perforate, is thus

$$\frac{j\omega m_I}{P} = \frac{j\rho ck}{P}\left[\tfrac{4}{3}\,l + \frac{16r_0}{3\pi}\bigg/\psi'(\xi)\right],$$

P being the porosity.

The specific acoustic reactance for the Helmholtz-type dependence is similarly given by (see equation (16))

$$\frac{j\rho ck}{P}\left\{l\left[1 + \frac{1}{r_0}\sqrt{\left(\frac{2\mu}{\rho\omega}\right)}\right] + \frac{16r_0}{3\pi}\bigg/\psi'(\xi)\right\}.$$

It is important to note that the resistive part of the specific acoustic impedance must be similarly modified. In practice the correction will be small for low porosity samples ($7\frac{1}{2}\%$) but for high porosity samples (22%) will reduce the end-correction by a significant amount.

The general equation for the normal specific acoustic impedance of a perforate of porosity P may therefore be written as

$$z = \frac{1}{P}\left[l_1(j\omega\rho)/\square(k_s r_0) + 16 r_0 (j\omega\rho)/3\pi\square(k_s' r_0)\,\psi'(\xi)\right], \tag{18}$$

for the case of Sivian's end-correction. A similar expression may be written based on the end-correction value experimentally determined by Ingard.

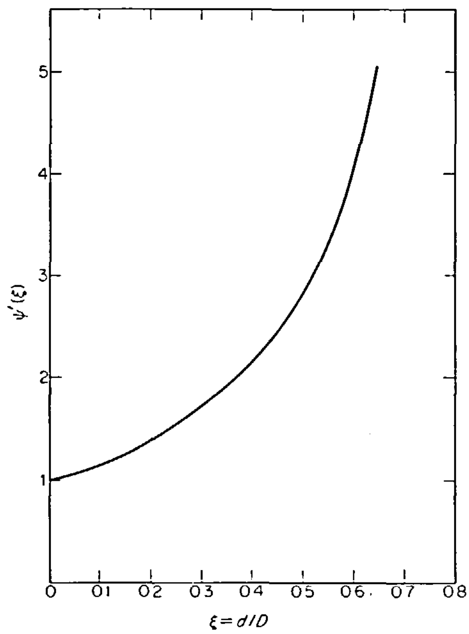

Figure 6. Fok's function $\psi'(\xi)$ *versus* $\xi(=d/D)$, $d = 2r$.

As implied by all the considerations outlined in this section, this formula, equation (18), is the best obtainable at present, on both theoretical and experimental grounds, for predicting the linear normal acoustic impedance of perforates, when radiation resistance is negligible.

3. MEASUREMENTS AND RESULTS

3.1. INTRODUCTION

This section is concerned primarily with experimental techniques, the measurement programmes and the results of the experimental programmes. Particular consideration is given to the dependence of the acoustic characteristics of the samples upon the parameters specifying the sample geometries and the disturbance amplitudes. Finally, some aspects of the work which, although of importance to the overall programme, do not have direct bearing on the prediction of the non-linear behaviour of the material, are also included.

As noted in section 1 the geometrical characteristics of the samples chosen were consistent with the requirements of those materials which might be employed in typical in-engine applications. The hole diameters chosen gave a range of 3:1 and, with the particular plate thicknesses selected, hole diameter to plate thickness ratios ranging from 1:1 to 5:1. This range in hole diameter also gave a reasonable variation in both Reynolds and Strouhal numbers (based on hole diameter) for the same plate porosity: i.e., 3:1. The range of plate thickness to hole diameter was adequate to illustrate the behaviour of the "end effects" or attached mass (see section 2) with sound pressure level. The range of variation in hole diameter to plate thickness was also adequate for determining the applicability of discharge coefficients to non-ideal orifices (see section 4).

Since interaction effects would depend on hole spacing, and hence on porosity, it was obviously desirable to vary this. A maximum acceptable porosity was considered to be 22% (ratio of open to total area). Three "standard" porosities were initially chosen: 22%, 15% and 7½%. Unfortunately, owing to difficulties with the manufacturers, it was not possible to obtain the 15% porosity sample. The tests were thus limited to the 22% and 7½% porosity samples.

The particular testing technique chosen—setting the backing cavity depth behind the sample equal to a quarter wavelength—ensured that the impedance measured was that due to the plate (this was considered as a more reliable technique than using an anechoic termination behind the sample—this aspect will be dealt with later). The maximum upper test frequency was limited by the cut-on of the (1, 0) mode in the tube, this being at approximately 3400 Hz. The lower test frequency was chosen to be consistent with that obtained from a 3 inch deep cavity behind the sample (i.e., approximately 1100 Hz). One further test frequency of approximately 1700 Hz was chosen, this being consistent with a 2 inch backing cavity. This range of frequencies thus gave a variation of approximately 1:1·7 in nominal resistance (Helmholtz controlled—see section 2) for a particular hole size, and a range of 3:1 in "effective" Strouhal number for a given hole and particle velocity. A comprehensive description of the samples is given in Table 1.

TABLE 1

Key to sample code—sample description†

Code used (after manufacturer‡)	Hole diameter (in)	Hole pitch (in)	Porosity (%)
RT 335	0·110	0·220	22·7
RT 153A	0·050	0·100	22·5
RT 233	0·026	0·052	22·6
RT 335/00	0·110	0·381	7·6
RT 153A/00	0·050	0·173	7·5
RT 233/00	0·026	0·090	7·5

† All samples were machined to a common thickness of 0·022 inches and hole pitch lines for all samples were 60°.

‡ Greenings Limited, Warrington.

The particular impedance measuring techniques employed (i.e., transmission line method and the two-microphone method) are dealt with separately in the following sections. However, before these methods and the results they produced are dealt with, the problems of the coupling of the mechanical and acoustic impedance of the sample will be examined, because any significant coupling would falsify the data, resulting in misleading conclusions for the interpretation of the material's acoustic behaviour.

3.2. THE COUPLING OF THE ACOUSTIC AND MECHANICAL IMPEDANCES OF THE TEST SAMPLE

Because of the low thickness of the test samples (perforated plate) consideration was given to the possible effects of the mechanical vibrations of the plate and the consequences of this on the measured acoustic impedance of the plate. The theoretical solution of this problem, even in simplified form, is complex (see reference [23]), and thus it was considered more expedient and conclusive if the problem and effect were to be examined experimentally. In principle, the solution of the problem is straightforward; since the mechanical and acoustic systems are acted upon by the same pressure the effective impedance is that given by the sum of the impedances of the two systems in parallel. Thus, providing the mechanical impedance can be defined, its influence on the measured acoustic impedance can be determined. The difficulty, of course, lies in determining the mechanical impedance. At best, it is only possible to ascribe an "average" impedance to the system. However, if the mechanical impedance is large compared with the acoustic impedance, then, by virtue of the shunt connection of it to the acoustic system, it will not affect the measured value of the acoustic impedance. Obviously the condition will not be met if the fundamental plate resonance is close to any of the proposed testing frequencies. Calculations of the plate fundamental frequency, for the assumed case of a non-porous "fixed–fixed" plate, indicated that the resonant frequency was of the order of two to three times that of the maximum test frequency envisaged in practice, indicating that at the planned test frequencies the mechanical system impedance was, at least, appreciably different from zero.

In order to further establish the significance, if any, of the mechanical system's impedance on the acoustic system, a series of measurements† were carried out when the samples were, respectively, both with and without honeycomb in the backing cavity behind the sample. In the former case (with the honeycomb) the sample is mechanically stiffened without significantly modifying the reactance of the cavity—i.e., the volume of the cavity taken up by the honeycomb could be regarded as negligibly small. The honeycomb consisted of $\frac{3}{8}$ inch hexagonal cells, the walls of which were constructed from 0·004 in thick aluminium foil. The depth of the honeycomb was made slightly larger than that of the backing cavity so that the sample was slightly pre-loaded.

Since the frequencies used in the measurement programme were below the estimated fundamental resonance of the plate it was therefore reasonable to assume that the impedance of the plate could be represented by a stiffness reactance. Sivian [10] has considered the problem of coupling of the mechanical and acoustic systems and the effect this has on the measured impedance. If the mechanical system has impedance $Z_2 = R_2 + j\chi_2$ and the acoustic system $Z_1 = R_1 + j\chi_1$ then, for the quarter wavelength condition, the measured impedance, Z, is given by the following expression [10]:

$$Z = R_1[\chi_2^2/R_1^2 + (\chi_1^2 + \chi_2^2)] + j\chi_1[1 + [(\chi_2/\chi_1)(R_1^2 + \chi_1^2) - (R_1^2 + \chi_1^2)]/R_1^2 + (\chi_1 + \chi_2)].$$

Thus, when $\chi_2 = -\infty$ (the reactance of the mechanical system is infinite), $Z = R_1 + j\chi_1$ and when χ_2 is finite errors are introduced into both R_1 and χ_1; in particular χ_1 is very sensitive to χ_2 having a finite value, whereas R_1 is relatively insensitive. The condition for smallness of error in the reactance is [10] $(-\chi_1\chi_2 + 3\chi_1^2) \gg R_1^2$. Thus, errors due to finite mechanical impedance should become more evident at high sound pressure levels since R_1 increases with it. By way of illustration [10], suppose $-\chi_2 = 9\chi_1$ and $R_1 = 3\chi_1$ (this latter equality is quite reasonable; see Figures 7 to 10) then $Z \doteq 1{\cdot}1\,R_1 + (-0{\cdot}12)j\chi_1$. Therefore, whilst there is only a 10% error in the measured resistance the reactance is very much in error; even the sign is wrong.

Examination of the data, in the light of the above comments, as will be demonstrated, reveals that the differences between the measurements with and without the honeycomb backing cannot be attributed to finite mechanical impedance of the sample, but rather the

† It is not proposed to describe the measurement technique in any detail here since it is essentially identical to that presented in section 3.4.

difference must be due to other effects. The results are presented and discussed in the following section.

3.2.1. *Experimental results and discussion*

Examples of the experimentally determined normal acoustic impedance of the six perforates both with and without the honeycomb backing are presented in Figures 7 to 12. Figures 7 to 9 and 10 to 12 are for 22% and $7\frac{1}{2}$% perforates, respectively. (The complete results are given in reference [16].) On each figure the real and imaginary part of the specific acoustic impedance ratio has been plotted against the free space sound pressure level (P_{INC}),

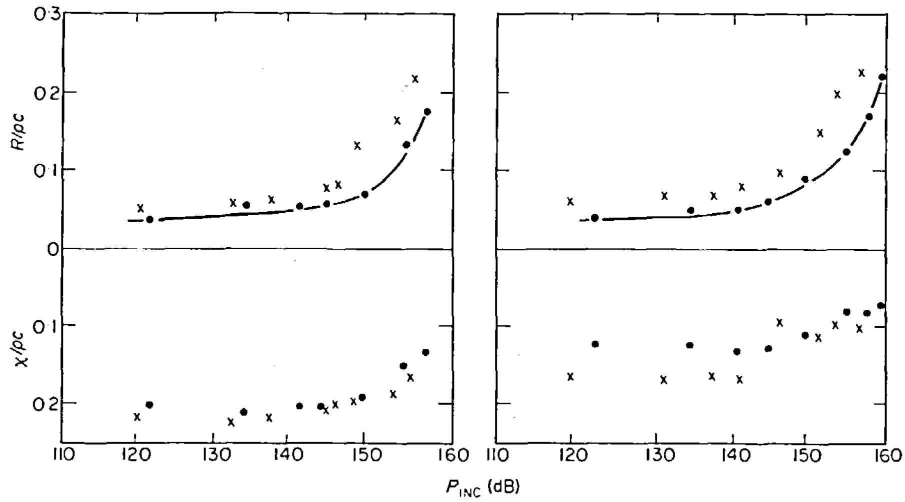

Figure 7. RT 335 (see Table 1). Figure 8. RT 153A (see Table 1).

Figures 7 and 8. Comparison of measured impedance with and without honeycomb backing; cavity depth 2 in ($\lambda/4 \simeq 1700$ Hz). $\times$, With honeycomb backing; $\bullet$, no honeycomb backing.

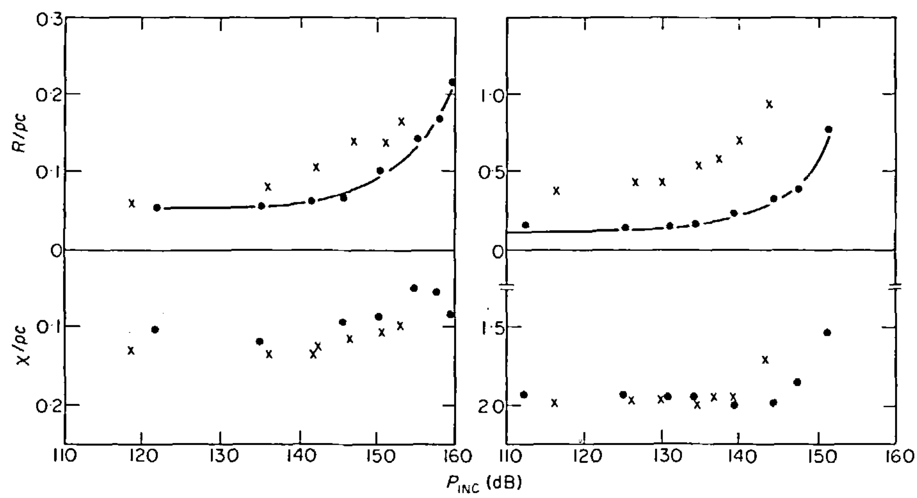

Figure 9. RT 233/00 (see Table 1). $\lambda/4 \simeq 1700$ Hz. Figure 10. RT 335/00 (see Table 1). $\lambda/4 \simeq 3400$ Hz.

Figures 9 and 10. Comparison of measured impedance with and without honeycomb backing; cavity depth 2 in. $\times$, With honeycomb backing; $\bullet$, no honeycomb backing.

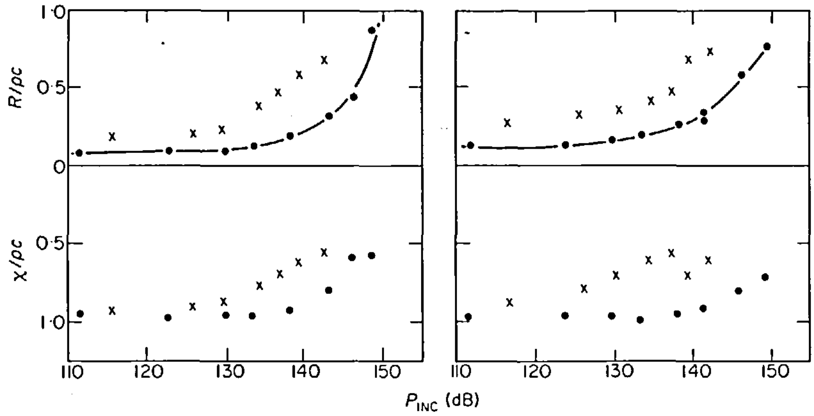

Figure 11. RT 153A/00 (see Table 1). Figure 12. RT 233/00 (see Table 1).

Figures 11 and 12. Comparison of measured impedance with and without honeycomb backing; cavity depth 1 in ($\lambda/4 \simeq 3400$ Hz). ×, With honeycomb backing; •, no honeycomb backing.

a smooth curve being drawn through the measured values of the specific acoustic resistance ratio for the measurements made without the honeycomb backing. Examination of all the results for the 22% perforate (of which Figures 7 to 9 are examples) shows that the measured mass reactance is essentially the same both with and without the honeycomb in the backing cavity. According to the previous discussion, if the mechanical impedance of the system (i.e., that due to plate vibration) were finite, its effect would be most noticeable on the measured reactance. Since no significant difference is evident between the honeycomb/no honeycomb results it is therefore reasonable to assume that the mechanical impedance is effectively infinite. The resistive impedance is, in all cases, and at all sound pressure levels, significantly larger for the honeycomb in the backing cavity. This aspect will be examined shortly.

The results for the $7\frac{1}{2}$% perforate (of which Figures 10 to 12 are examples) indicate—with the exception of two sets of data (not shown here)—the same general behaviour and hence the same conclusions may be drawn. The exceptional cases may be attributed to experimental error (the reactive impedance is very sensitive to errors in minimum location).

The resistive impedance is significantly different (larger) with the honeycomb in the backing cavity in all cases, especially at high frequencies, but particularly in the case of the 22% perforates. This increase in the resistive impedance is not consistent with possible changes resulting from the coupling of the mechanical impedance of the sample to the acoustic impedance: were this the case, then the reactance would also have been significantly different in the two cases. For example, if the resistance with the honeycomb is taken to be due to mechanical/acoustic impedance coupling, then if the uncoupled impedance were to be twice the coupled impedance—with the decoupled reactance of the order of ten times the resistance—it would be necessary that the measured reactance for the coupled system should be some three times in magnitude the reactance of the uncoupled system and of the opposite sign; clearly (see, for example, Figure 11) this is not the case. Whilst the results noted above provide sufficient grounds for not using the honeycomb in the backing cavity it is of considerable interest to establish the reason for the difference in measured resistances in the two cases.

As noted earlier, the honeycomb in the backing cavity generally results in a resistive impedance which is larger than that for the sample alone. It is convenient to examine the results from the two porosities separately. Comparison of the results for the 22% open area perforates (see, e.g., Figures 7 to 9) with and without the honeycomb in the backing cavity indicate that with the honeycomb in the cavity (a) the nominal resistance value is higher and (b) the slope of the non-linear region is similar. The increase in nominal resistance is to be anticipated since in placing the honeycomb in the backing cavity more surface area has been introduced and hence more viscous losses will result. A quantitative estimate of the losses can be readily obtained by replacing k in the usual expression for the impedance of a lossless cavity by a complex propagation constant, where the attenuation coefficient is the usual Helmholtz form (viz., $|(\omega\mu/2\rho)^{1/2}/rc|$). By extracting the real part, it can be shown that, to a first approximation, the specific acoustic resistance ratio due to the viscous losses resulting from the honeycomb are approximately $R_{\mathrm{H}} \simeq [\sinh(\sqrt{\omega\mu/2\rho}/rc)d]/\pi r^2$, where r is the radius of the honeycomb cells, the other symbols having the usual meanings. For convenience it has been assumed that the hexagonal honeycomb can be represented by a circular one of diameter equal to the hexagonal cell size. This expression gives a nominal value for the specific acoustic impedance ratio of the honeycomb at 3400 Hz of 0·015. Since the measurements are carried out for quarter wavelength cavity depths the nominal honeycomb resistances at the different cavity depths (and hence frequencies) are proportional to the square root of the cavity depth (the function being approximately linear). Hence,

$$R_{\mathrm{H}_{3400}}/R_{\mathrm{H}_{1130}} = \frac{\text{cavity depth at 3400 Hz}}{\text{cavity depth at 1130 Hz}} = \frac{\sqrt{1}}{\sqrt{3}},$$

or $R_{\mathrm{H}}(3400\ \mathrm{Hz}) \times 1{\cdot}7 = R_{\mathrm{H}}(1130\ \mathrm{Hz})$. Therefore the nominal resistance due to the honeycomb is higher at the larger cavity depth (or lower frequencies). The estimated nominal specific acoustic resistance ratios due to 3 inch, 2 inch and 1 inch cavities are, respectively, 0·25, 0·21 and 0·15.

The measured resistance, at low sound pressure levels, will be the sum of the perforate resistance and the honeycomb resistance since the volume velocity is continuous. Alternatively, the measured resistance with the honeycomb in the backing cavity should exceed the resistance for the plain cavity configuration by an amount equal to the nominal resistance of the honeycomb. Comparison of the experimental and theoretical results gives satisfactory agreement between predicted and observed trends; for the 3 inch cavity the measured results for the specific acoustic resistance ratios are in the range 0·02 to 0·03, the calculated value being 0·025; for the 2 inch cavity the experimental results are in the range 0·02 to 0·025, the calculated result being 0·021; for the 1 inch cavity the measured results lie in the range 0·03 to 0·05, compared with a calculated value of 0·015. The difference in this last case is significant and it is also of interest to note that the largest deviation occurs with the largest hole size. This aspect of the behaviour will be referred to later.

It was also noted earlier in the text that with the honeycomb present the non-linear region commenced at a lower sound pressure level than for the plain cavity case. This aspect of the behaviour is reasonable, because of the following considerations. Since the non-linear resistance is velocity dependent (see section 4) and also since the results are plotted against a free space travelling pressure (i.e., the pressure which would be seen with an anechoic termination) then, for a given pressure, P_{INC}, the two configurations will experience different velocities. The modulus of the velocity is given by $|v| = (|P|R)/(R^2 + \chi^2)^{1/2}$. Taking mean values for the resistance and reactance in the two configurations of R(perforate) $= 0{\cdot}04$, R(honeycomb) $= 0{\cdot}02$ and χ(all configurations) $= 0{\cdot}15$ then gives v(perforate)/v(perforate + honeycomb) $= 0{\cdot}04/0{\cdot}06$: i.e., for a given P_{INC} the honeycomb configuration velocity is 1·5 times the plain cavity condition velocity. Notwithstanding this, the two measured results should converge at high sound pressure levels where the non-linear resistance dominates. The 3 inch cavity results (where, in general, the highest non-linear resistance is exhibited) tend to indicate a convergence; however, the ratio of the nominal to non-linear values is only of the order of some 4:1 (for the honeycomb in the backing cavity).

The $7\frac{1}{2}\%$ perforates, except, again, for the 1 inch cavity results (and within the limits of the possible scatter due to experimental error) fit the present model well. For these perforates the nominal resistance is much higher (of the order, approximately, of 0·1); thus the honeycomb resistance is much lower in comparison and the shift in the two curves is much less evident. Indeed, at high sound pressure levels the non-linear resistance curves, as was to be expected, merge. This statement is, of course, not true for the 1 inch cavity results (see, e.g., Figures 10 to 12). Both for the 22% and $7\frac{1}{2}\%$ perforates the 1 inch cavity (i.e., the high frequency) results do not satisfactorily fit the present model, at least for the nominal resistance, and it is also noteworthy that in both cases, the most significant differences between the results for the two configurations are for the largest hole size (RT 335, hole diameter = 0·110 inch; see, e.g., Figures 7 and 10). Whilst, for these results, the slopes of the non-linear regions are, in general, similar and hence the differences between results for the two configurations can be described as a shift along the abscissa, the amount of shift—which is (according to the model presented) dependent on the nominal resistance—cannot be explained in quantitative terms: i.e., the apparent resistance due to the honeycomb in all cases is far in excess of the calculated value. No satisfactory explanation can be given to this; it can only be said that it is some frequency dependent effect which manifests itself only above a certain frequency and is peculiar to the configuration with the honeycomb in the backing cavity.

In conclusion, it has been demonstrated by both experimentally determined results and simplified theoretical models that the coupling of the mechanical and acoustic impedances of the sample can be disregarded. On the other hand it has been demonstrated that the honeycomb in the backing cavity has a positive effect on the measured resistance. This is most evident for samples of low nominal resistance (e.g., 22% perforates) but is particularly evident in the samples examined here at the high frequencies.

3.3. THE TWO-MICROPHONE METHOD OF IMPEDANCE MEASUREMENT

3.3.1. *Introduction*

This method of impedance determination and the theoretical background have been described in references [15] and [16]. Considerable investigation of this method of impedance determination was carried out because, as noted earlier, it provides a very convenient and rapid method as compared with the standard transmission line technique. The impedance determinations by this technique were carried out at the same time as those using the transmission line technique, thus enabling direct comparison of results to be made: i.e., effects due to temperature and geometrical variation between consecutive tests are thereby eliminated.

Further, the second microphone, located in the back wall of the backing cavity, gives (see reference [15]) a direct measure of the acoustic particle velocity independent of derived values such as the impedance of the sample. It should be noted that the velocity determined by this method is dependent on the pressure calibration of the microphone (i.e., at worst ± 1 dB) and the accurate determination of the cavity depth. Sensitivity to the latter aspect is negligible.

The difficulty in practical application of the technique arises from selecting the correct microphone location for determining the pressures and phase at the front face of the sample. This difficulty, which is apparent from qualitative appraisal of the acoustic field in the vicinity of the sample, is evident from the results obtained.

3.3.2. *Experimental configuration and programme; results*

The experimental configuration is identical to that used for the standard transmission line technique with the addition of a second microphone mounted in the end wall of the sample backing cavity; in these tests a $\frac{1}{4}$ inch Brüel and Kjær condenser microphone was used. In addition to the "standard" equipment, a microphone switching unit and a phase meter were also used in order that the phase and pressure from the two microphones could be conveniently measured. The switching unit was a Brüel and Kjær Type 4408 and the phase meter was the ADYU Type 406L. The set up is shown schematically in Figure 13.

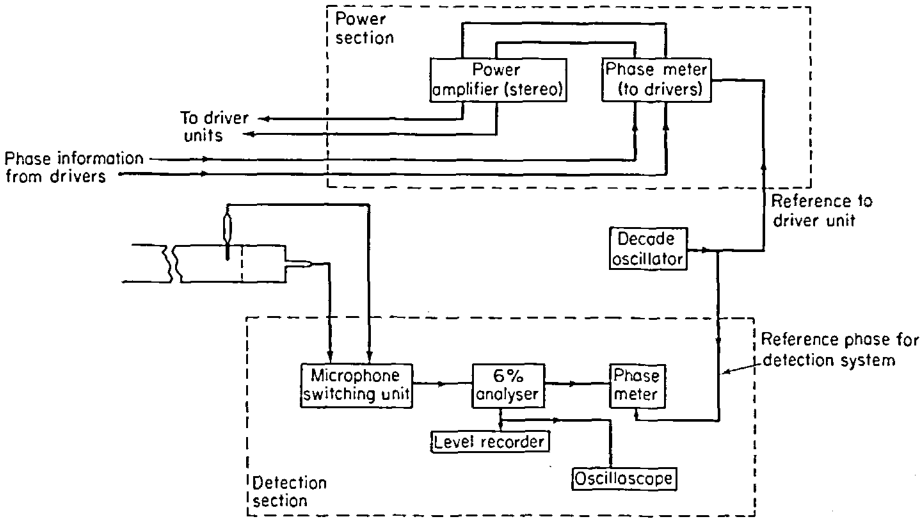

Figure 13. Schematic arrangement of the experimental set up used in the two-microphone method for impedance determination.

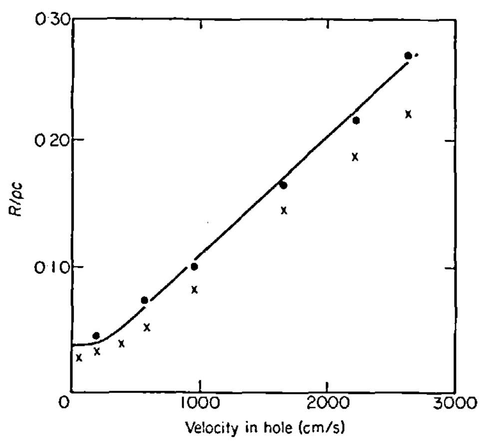

Figure 14. Comparison of measured resistive impedance using transmission line and two-microphone techniques for RT 153A†, cavity depth 3in ($\lambda/4$ at 1130 Hz). ●, Transmission line method; ×, two-microphone method. †See Table 1.

Since it was not possible to determine directly the phase between the two microphones (only one signal could be fed into the analyzer and hence into the phase meter) each one was compared with a reference signal derived from the decade oscillator. In order to determine the absolute phase difference between the two microphones a technique identical to that for pressure calibration was used; the microphones were brought into close proximity and the necessary readings taken (see reference [15]). In practice it was found necessary to calibrate the microphones for phase at the beginning and end of each impedance determination.

Further, it was found that the phase calibration was also sound pressure sensitive, presumably a result of disturbing the microphone's attached mass.

The procedure for carrying out impedance determinations was therefore as follows: (a) calibrate microphones; (b) carry out measurement both for the two microphone and transmission line techniques; (c) recalibrate microphones.

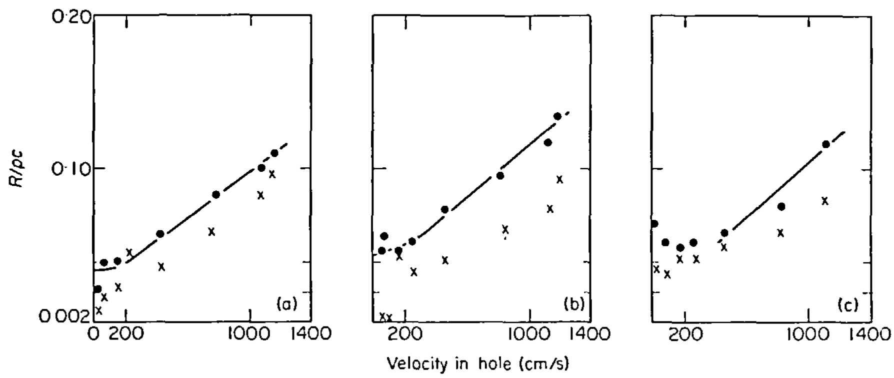

Figure 15. Comparison of measured resistive impedance using transmission line and two-microphone techniques for the samples (a) RT 335, (b) RT 153A and (c) RT 233†, cavity depth 1 in ($\lambda/4$ at 3400 Hz). ●, Transmission line method; ×, two-microphone method. †See Table 1.

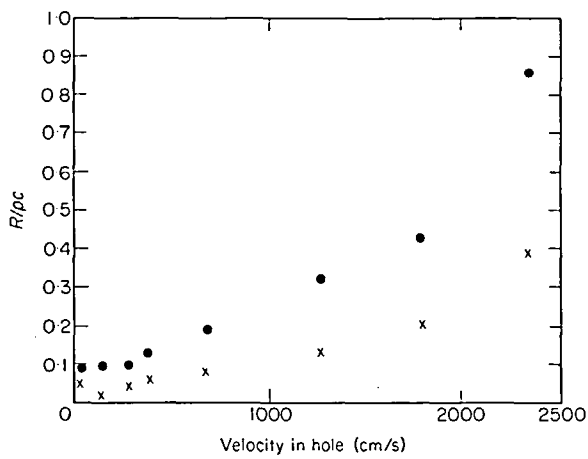

Figure 16. Comparison of measured resistive impedance using transmission line and two-microphone techniques RT 153A/00†, cavity depth 1 in ($\lambda/4$ at 3400 Hz). ●, Transmission line method; ×, two-microphone method. †See Table 1.

The data was subsequently processed, by using a digital computer, to yield the real and imaginary parts of the specific acoustic impedance and also the acoustic particle velocity. The computer programme is elementary and has not been included. Several representative results, for the specific acoustic resistance ratio of the samples, are presented in Figures 14 to 16. The results for specific acoustic mass reactance ratio are presented in tabular form (Tables 2 and 3) since graphical presentation, in this instance, would not provide significantly more assistance in data interpretation.

Figure 17 shows both the relative phase difference between the two microphones on subsequent occasions, together with the phase variation with distance along the tube, the distance being measured from the sample face towards the source end of the tube.

TABLE 2

Comparison of measured specific acoustic reactance ratio of liner for 22% perforates using two-microphone and transmission line techniques†

3 inch cavity		2 inch cavity		1 inch cavity	
2-microphone method	Transmission line method	2-microphone method	Transmission line method	2-microphone method	Transmission line method
		RT 335			
0·15	0·15	0·22	0·20	0·43	0·47
0·15	0·16	0·23	0·21	0·42	0·46
0·15	0·13	0·23	0·21	0·39	0·44
0·14	0·13	0·23	0·21	0·43	0·48
0·12	0·11	0·22	0·19	0·43	0·46
0·94	0·07	0·19	0·15	0·42	0·46
0·91	0·08	0·22	0·13	0·36	0·43
0·96	0·09	0·17	—	0·43	0·42
		RT 233			
0·09	0·09	0·13	0·16	0·24	0·26
0·09	0·09	0·13	0·12	0·24	0·25
0·08	0·07	0·12	—	0·24	0·25
0·08	0·07	0·12	0·10	0·24	0·26
0·07	0·05	0·11	0·09	0·24	0·25
0·07	0·06	0·10	0·05	0·23	0·23
0·08	0·05	0·95	0·06	0·21	0·19
0·08	0·04	0·82	0·08	0·20	—
		RT 153A			
0·11	0·09	0·16	0·13	0·10	0·31
0·10	0·10	0·16	0·13	0·30	0·29
0·10	0·10	0·16	0·14	0·29	0·31
0·09	0·09	0·15	0·13	0·30	0·31
0·08	0·08	0·13	0·12	0·29	0·28
0·10	0·06	0·11	0·09	0·29	0·27
0·08	0·07	0·10	0·09	0·27	0·27
0·09	0·05	0·09	0·08	0·28	0·27

† Each pair of results is for the same in-hole velocity.

3.3.3. *Discussion of results*

It is convenient, at least initially, to consider the results for the two porosities separately.

3.3.3.1. *The 22% porosity perforates.* The real and imaginary parts of the specific acoustic impedance ratio are discussed separately below.

(i) *Specific acoustic resistance ratio.* Some results are given in Figures 14 and 15, the largest cavity results being presented first. Several trends are immediately apparent: (a) the resistance measured by the two-microphone method is always lower than that measured by the transmission line method; (b) the slope of the non-linear region is, in general, the same when measured by either technique; (c) there is no consistent trend for the observed "abscissa shifts", or slope off-set of the results for different samples and different cavity depths.

(ii) *Specific acoustic reactance ratio.* The results, presented in Table 2, do not indicate any one specific trend; however, the following general trends may be observed: (a) the measured values for both methods corresponding to the lower sound pressure levels show good agreement for all sets (cavity depths) of data but particularly for the 3 inch cavity results; (b) the

TABLE 3

Comparison of measured specific acoustic reactance ratio of liner for 7½% perforates using two-microphone and transmission line techniques†

3 inch cavity		2 inch cavity		1 inch cavity	
2-microphone method	Transmission line method	2-microphone method	Transmission line method	2-microphone method	Transmission line method
		RT 335/00			
0·60	0·49	0·1004	0·73	1·09	1·96
0·55	0·51	0·99	0·71	1·08	1·96
0·45	0·42	0·99	0·71	1·08	1·98
0·38	0·36	0·10	0·71	1·12	1·97
0·26	0·35	0·77	0·66	1·00	2·02
0·43	0·39	0·55	0·54	1·09	2·00
0·38	0·47	0·57	0·37	1·06	1·87
0·45	0·42	0·62	0·35	0·89	1·55
		RT 233/00			
0·29	0·28	0·50	0·47	0·59	0·99
0·24	0·26	0·49	0·45	0·57	0·96
0·22	0·29	0·44	0·39	0·57	0·97
0·29	0·24	0·40	0·37	0·57	1·00
0·21	0·29	0·37	0·32	0·53	0·96
0·30	0·28	0·24	0·32	0·48	0·93
0·33	0·27	0·30	—	0·49	0·82
0·33	0·29	0·29	—	0·52	0·72
		RT 153A/00			
0·30	0·3	0·47	0·47	0·58	0·96
0·27	0·26	0·47	0·46	0·56	0·98
0·24	0·20	0·43	0·40	0·57	0·97
0·20	0·18	0·38	0·36	0·57	0·98
0·22	0·11	0·38	0·17	0·56	0·94
0·25	0·24	0·32	0·25	0·47	0·80
0·28	0·28	0·31	—	0·46	0·60
0·30	—	0·30	—	0·48	0·58

† Each pair of results is for the same in-hole velocity.

two-microphone results are higher at the higher sound pressure levels for the 3 inch and 2 inch cavity results; (c) in general, the two-microphone results are lower for the 1 inch cavity case. The differences are, however, in general, within the bounds of the experimental errors of the two methods. The correlation between results is generally good with the exception of those for the 3 inch cavity and the sample RT 335 (having the largest holes).

The conclusion drawn from the theoretical analysis presented in reference [15] was that if either the pressure at the front or back face were incorrect (it being assumed, for the moment, that the phase angle is correct) this would result in the resistance and reactance both being either too large or too small. For these particular results the two-microphone resistance is always too small; this implies that the measured pressure at the front face is too small (the pressure at the back is taken to be correct). Where this is the case the measured mass reactance values (by the two-microphone method) would also have to indicate a tendency towards being too small. The results clearly indicate the converse of this. It was noted in reference [15] that the resistance value was more sensitive to errors in phase angle.

Further, because of the identity $\cos^2\theta + \sin^2\theta = 1$, it is apparent that if the value for phase is too low the measured resistance will also be too low whereas the mass reactance value will be too high (resistance is dependent on the sine and mass reactance is dependent on the cosine of the phase angle). Thus it would appear that the error lies in the determined phase angle. However, before commenting further, the results for the other porosity will be examined.

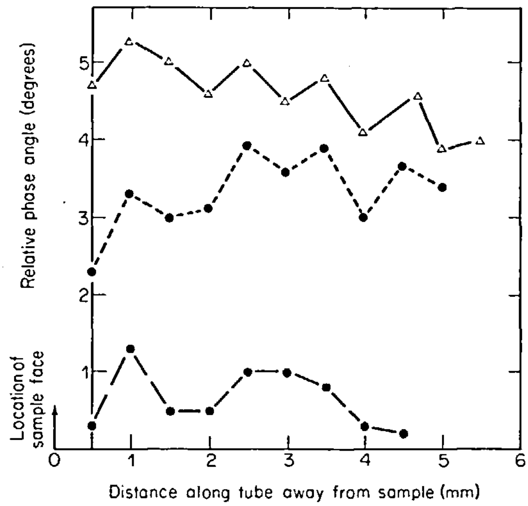

Figure 17. Phase angle relative to a reference for various locations along the tube taken on subsequent occasions frequency 1700 Hz ($\lambda/4 \simeq 2$ in). Note: probe microphone parallel to sample face. Axis of tube coincident with open end of probe. Probe diameter 14 G.

3.3.3.2. *The $7\frac{1}{2}$% porosity perforates.* The real and imaginary parts of the specific acoustic impedance are discussed separately under subsections (i) and (ii) below.

(i) *Specific acoustic resistance ratio.* One of the measured results is shown in Figure 16. The results as a whole (see reference [16]) follow the same trends as those for the 22% perforates except that the scatter is, in general, much reduced, with the exception of the results shown in Figure 16 for the 1 inch cavity. The general conclusions which may be drawn from the results are, therefore, the following: (a) the resistance measured by the two-microphone method is generally lower than that measured by the transmission line method; (b) the slope of the non-linear region (resistance *versus* "in hole" velocity) is the same; (c) there is no consistent mean off-set between the slopes in the non-linear region.

(ii) *The specific acoustic reactance ratio.* The results are presented in Table 3. The trends are similar to those for the 22% perforate, with the exception of those for the 1 inch cavity where the two-microphone results are considerably less than those determined by the transmission line method. Therefore, if for the present, one disregards the 1 inch cavity results, the general conclusion is that the two-microphone results are greater than those determined by the transmission line method.

As the results for the $7\frac{1}{2}$% perforates are in line with the trends observed for the 22% perforates (but to a lesser extent), similar conclusions may be reached about the explanation of the difference in results in this case: i.e., that phase angle errors are giving rise to the differences between the two sets of experimentally determined results. On the basis of this assumption, a limited investigation into the behaviour of the phase angle, both with distance and time, at the sample face was carried out. The results are presented in Figure 17. Several interesting aspects of the behaviour are apparent: (a) compared with that of the reference signal, the phase of the two microphones could drift by up to 5 degrees; (b) for points approaching the sample face, the phase is changing rapidly with distance; (c) on all occasions the spatial variations of phase were very similar.

In practice the measured phase angle has a considerable range, depending on the cavity depth (the value given in reference [15] was only a broad-based average); consequently, the errors involved depend upon the cavity depth. Correspondingly, in Table 4 the errors have been computed for both sample porosities by using a mean phase angle (over the measured range of values) and with an assumed error in measurement of 2 degrees—composed, jointly, of a "drift" and measuring error.

TABLE 4

Errors in resistive and reactive components of impedance for mean phase angle at various cavity depths

Cavity depth (in)	Range of phase angle (deg)	Mean	% error $(\theta - 2)°$ $R/\rho c$	$\chi/\rho c$	% error $(\theta + 2)°$ $R/\rho c$	$\chi/\rho c$	Total error $R/\rho c$	$\chi/\rho c$	Porosity
1	26–30	28	−6·6	+1·8	+6·6	+1·8	13·0	4·0	7½%
2	71–84	77	−1·0	+14·0	+7·0	−15·5	2·0	30·0	7½%
3	17–72	45	−3·5	+3·5	+3·4	−3·4	7·0	7·0	7½%
1	9–21	15	−13·5	+9·5	+24·0	−2·0	24·0	2·0	22%
2	11–59	35	−5·1	+2·3	+10·0	−5·0	10·0	5·0	22%
3	15–70	37	−4·6	+2·6	+4·7	−2·6	9·5	5·5	22%

The results in Table 4 reflect the general trend in the measured results: i.e., that of too low a resistance value and too high a mass reactance value for a negative error in phase angle measurement. In other words, the phase angle determined is too low. It is apparent, as was to be expected, that the error is dependent on the phase angle; further, considerable errors in impedance determination can result from errors in phase determination. In these particular calculations phase errors of 2 degrees were assumed; however, as illustrated in Figure 17, errors of 5 degrees or more may occur, these presumably being a result of "drift" in some component of the equipment. For example, if a 5 degree (negative) error is assumed in the 1 inch cavity 22% perforate phase angle determination, this would result in a 33% error in the resistance determination. Because of the lack of consistency in the off-set of the slopes of the non-linear regions between both sets of results, it is suspected that the errors are drift-derived rather than systematic. Further, this difficulty, illustrated in Figure 17, was noted during the measurement programme whilst phase-calibrating the microphones; a mean phase difference taken from the results at the beginning and end of the measurements was taken as the best value. A further, although apparently second-order error, is also apparent from Figure 17; as the sample face is approached the phase angle is seen to be decreasing (in all the presented cases), although it is not apparent from the results (other than by extrapolating back through two points to the sample face) what the phase angle at the face is or should be taken to be. There appears to be, therefore, a problem of defining exactly where phase should be determined.

The results therefore indicate that, at least for the present, the major difficulty lies with the instrumentation for phase measurement. Such drift could not have been attributed to temperature fluctuation in the tube because of both the time which is taken to carry out a measurement run and the thermal capacity of the facility. It should also be pointed out that the equipment used was supposed to have very good phase stability.

In comparison, and because of the preceding comments, pressure determination errors appear insignificant, but undoubtedly this aspect also warrants closer investigation.

In conclusion, it would appear that before detailed consideration can be given to the two-microphone method of impedance measurement instrumentation difficulties need ironing out. Other effects such as phase, pressure and the correct location for determining these in relation to the sample face can then be examined in detail.†

3.4. TRANSMISSION LINE METHOD OF IMPEDANCE MEASUREMENT

3.4.1. *Introduction*

The general principles and method of extraction of impedance values from the results for the transmission line method of impedance determination have been described in reference [15]. The method is reliable in comparison with other methods, particularly when measuring non-linear impedance, since it does not rely on any physical measurements close to the sample face. However, it does have its limitations: in particular, for samples with low standing wave ratios, the determination of the minima becomes difficult even if a linear rather than logarithmic print-out of the standing wave pattern is used. Notwithstanding this, the method does permit a high degree of confidence to be placed in the results it produces; further the errors and sources of errors are well appreciated.

As noted previously, for this particular programme of measurements the sample was mounted such that the backing cavity depth represented a quarter wavelength for the particular frequency under test. This method was selected in preference to a fixed cavity depth technique because of the need with the latter to subtract the cavity impedance from the results and also because of the reduction in particle velocity this cavity impedance would have resulted in at some test frequencies. An alternative solution would have been to use an anechoic termination; however, whilst in principle this is an ideal solution, in practice it is difficult to obtain for the required ranges of frequency and sound pressure levels (the latter giving rise to non-linear impedance changes in the termination). Consequently, even though more work was involved, it was concluded that the quarter wavelength configuration was the most suitable and dependable.

Considerable care was taken to ensure that the sample was accurately mounted and that the edges, where the sample seated on the fixing edges in the tube, was airtight. In order to ensure an airtight seal the edges of the sample under test were coated with a very thin rubber solution; thus, when the sample was sandwiched between the seating edges of the tube and the "clamping box", all possible leakage paths were sealed.

3.4.2. *The programme of measurements*

With the facility calibrated, extraction of the data from the measurements was a routine procedure. In practice each sample was mounted in the tube and for a given backing cavity at least two independent sets of determinations carried out. Each sample was measured in turn for a given cavity depth; the cavity depth was then changed and the process repeated. Eighteen combinations of cavity depths and samples were measured and at least six different sound pressure levels were used. For each test condition, as discussed earlier, measurements were also taken to enable the impedance to be determined by the two-microphone method. The maximum sound pressure levels at which the tests were carried out were determined jointly by the cavity depth and the "aging" of the driver unit coils. Presumably at the smaller backing depths the length of the section was such as to present a considerable load impedance to the drivers, resulting in lower particle velocities in the samples. Similarly, as the units

† Subsequently to this work, the two-microphone method has been developed for *in situ* impedance measurements in flow ducts by Dean [5].

aged, the diaphragms tended to soften; this resulted in a lower output from them for a given current through them. For all the perforates tested the maximum attainable sound pressure levels (and hence acoustic particle velocities) were for the 3 inch backing cavity. The results are presented and discussed in the following section.

3.4.3. *Experimental results*

As has been previously noted (section 2 and reference [15]), the impedance can be plotted against several parameters. These include P_{INC} (the free space travelling pressure), the particle velocity in the hole, the volume velocity, or the total pressure at the sample face; other parameters such as Reynolds number and Strouhal number may also be used.

In anticipation of the results of the analysis in section 4, and from physical arguments, a dependence which is of considerable interest is that of the impedance on the particle velocity in the hole. As alternative parameters, the Reynolds number, Strouhal number or the volume velocity are generally simply scaled versions of this parameter. Therefore, the impedance plotted as a function of velocity in the hole is of considerable interest. (The usefulness of the

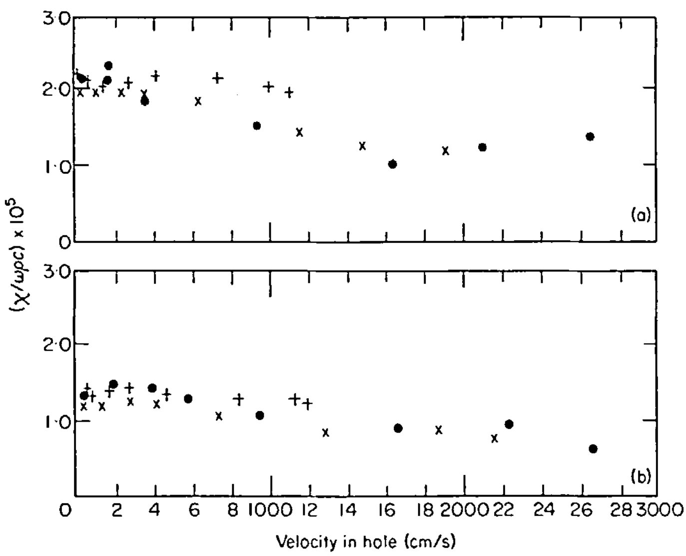

Figure 18. Measured reactance (transmission line method) plotted against velocity in hole; (a) RT 335, (b) RT 153A. ●, 3 in cavity; ×, 2 in cavity; +, 1 in cavity.

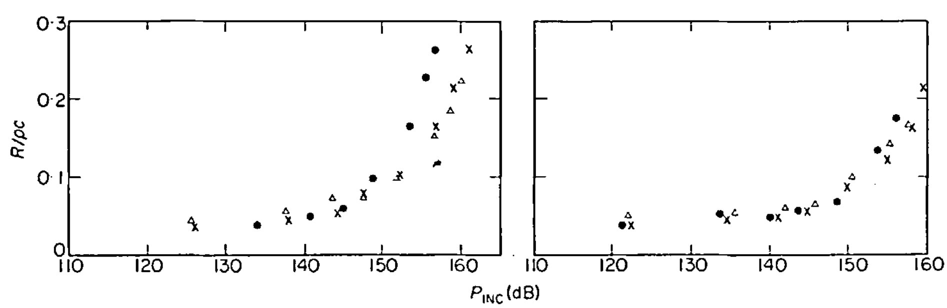

Figure 19. Backing cavity 3 in, $\lambda/4 \simeq 1130$ Hz. Figure 20. Backing cavity 2 in, $\lambda/4 \simeq 1700$ Hz.

Figures 19 and 20. Measured resistive impedance for the 22% perforates ●, RT 335; ×, RT 153A; △, RT 233.

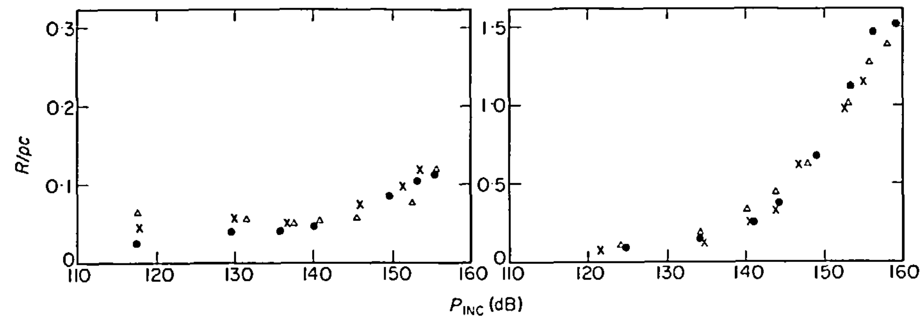

Figure 21. Backing cavity 1 in, 22% perforates, $\lambda/4 \simeq 3400$ Hz.

Figure 22. Backing cavity 3 in, $7\frac{1}{2}$% perforates, $\lambda/4 \simeq 1130$ Hz.

Figures 21 and 22. Measured resistive impedances. ●, RT 335/00; ×, RT 153A/00; △, RT 233/00.

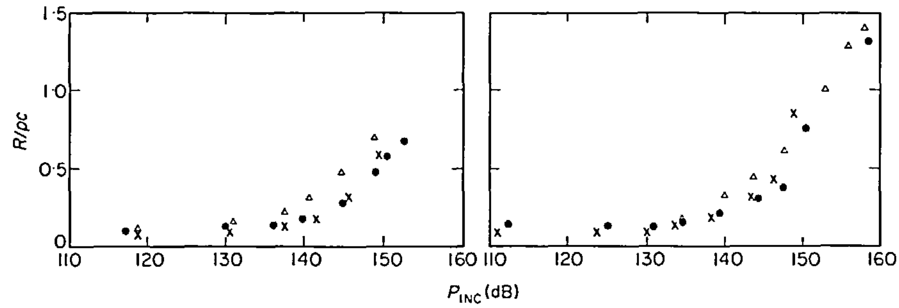

Figure 23. Backing cavity 2 in, $\lambda/4 \simeq 1700$ Hz.

Figure 24. Backing cavity 1 in, $\lambda/4 \simeq 3400$ Hz.

Figures 23 and 24. Measured resistive impedance of the $7\frac{1}{2}$% perforates. ●, RT 335/00; ×, RT 153A/00; △, RT 233/00.

other parameters is somewhat limited in comparison.) Consequently, the results have been presented as functions of velocity. Typical samples of the variation of the specific acoustic resistance ratio as a function of "in-hole" velocity have already been presented in Figures 14 to 16. An example of variation of the specific acoustic mass reactance ratio with "in-hole" velocity is given in Figure 18; for these cases the results for the same sample for each cavity condition have been collapsed on to one graph (N.B.: collapsed resistance data will be shown and discussed in section 4). Examples of the data for the specific acoustic resistance and reactance as functions of P_{INC} also have already been presented in Figures 7 to 12. For convenience, all the resistive impedance values for the different perforates having the same porosity and at the same cavity depth (test frequency) are shown plotted against P_{INC} in Figures 19 to 24.

3.4.4. *Brief discussion of results; errors*

It is convenient at this stage to discuss certain aspects of the results—in particular, the results plotted against P_{INC}—and to comment in general terms on the behaviour of the impedance with the in-hole velocity.

The usefulness of the P_{INC} plots is limited (from a theoretical analysis standpoint) to determining the nominal values of the impedance: i.e., those values exhibited at low sound pressure levels. This, in particular, is a reason for their inclusion; the non-linear behaviour of the mass reactance is also more evident when plotted against this parameter rather than velocity; the knee, or break point, tends to be sharper.

In section 2 a semi-empirical model was presented for calculating the nominal impedance when interaction effects between the holes occurred. Similarly, this same model can be used to calculate the nominal impedance for the case where interaction effects are neglected; this is achieved by putting the Fok function ($\psi'(\xi)$) equal to unity. However, before proceeding further with this discussion, several aspects of the assumptions on which the application of the Fok function is based should be emphasized.

The Fok function is applicable only to those conditions where potential flow conditions apply; clearly, in the non-linear region where turbulent flows exist interaction effects may be controlled by other effects. For example, it might be expected that for resistive interaction the controlling factor would be the rate of spread of the turbulent jet. In the case of the mass reactance, since it is essentially a pressure gradient effect and because the pressure gradient will not be too perturbed for "moderate" turbulence, it would be expected that the Fok interaction correction would be more applicable at higher velocities to the mass reactance term than to the resistive one.

This trend is indicated in the results of Figures 7 to 12; the mass reactance tends to be constant, in general, for P_{INC} less than some 135 dB, whereas, in the majority of cases, the resistive impedance, as far as is evident from the results, is still tending to a lower, presumably asymptotic, value. Unfortunately, owing to limitations with the power amplifiers, the impedance could not, at the time of measurement, be measured at lower levels than those values presented.

In view of these considerations a valuable comparison of the calculated nominal values of impedance and the lowest measured values can be made.

In Table 5 a comparison is made between the calculated nominal values for mass reactance and the values obtained from the experimentally determined results for the lowest velocities through the holes. The agreement between calculated and experimental results is very good when the Fok function is used; this is particularly noticeable in the case of RT 335 (0·110 inch hole diameter), where the reactance is controlled by the end effect. If, in this case, interaction effects are neglected, the calculated value is in error by as much as 100%. This error decreases, as would be expected, with decreasing hole size, since the end correction controls the mass reactance to a lesser extent.

A comparison of the calculated nominal resistance and measured lowest resistance value is made in Table 6. Clearly the agreement is poor between the interaction corrected (*via* Fok's function) value but good between the measured value and the calculated value when no interaction is assumed. This poor agreement between the calculated value, "corrected" for interaction, and the measured value is most likely due to non-linear behaviour. The flow through the hole is not a potential flow and therefore the application of the Fok function is invalid. The agreement with the values calculated when no interaction is assumed is probably fortuitous, since, from the measured data (see, e.g., Figures 7 to 12), it is indicated that the resistance has not reached its lower limiting value.

The Fok-based interaction correction appears to give good agreement with experimental results for the reactive component of the impedance. The difference in the resistive components is thought to be a result of the higher sensitivity of this component of the impedance to turbulent flow in the region of the orifice, which invalidates the application of the Fok correction method.

Examination of all the measured results when plotted against velocity (see, e.g., Figures 14 to 16) reveals good linear correlation between the velocity in the hole and the specific acoustic resistance ratio; in general the linearity is good above a velocity of 1000 cm/s. The lines have not been drawn in Figure 16, for example, because of insufficient data points at the higher velocities.

The results for the specific acoustic mass reactance (see, e.g., Figure 18) indicate that the

TABLE 5

Comparison of calculated and measured nominal values of $\chi/\omega\rho c$ (CGS units) for the two perforate porosities: $\psi' = 1$ no interaction effects; $\psi' \neq 1$ interaction included

(N.B.: see Table 1 for comprehensive description of samples)

	Sample description					
Porosity 22%	RT 335 Calculated	RT 335 Measured	RT 153A Calculated	RT 153A Measured	RT 233 Calculated	RT 233 Measured
$\psi'(\xi) = 1$	$3{\cdot}8 \times 10^{-5}$	$1{\cdot}9$–$2{\cdot}3 \times 10^{-5}$	$2{\cdot}14 \times 10^{-5}$	$1{\cdot}2$–$1{\cdot}45 \times 10^{-5}$	$1{\cdot}46 \times 10^{-5}$	$1{\cdot}0$–$1{\cdot}3 \times 10^{-5}$
$\psi'(\xi) = 2{\cdot}95$	$1{\cdot}77 \times 10^{-5}$		$1{\cdot}21 \times 10^{-5}$		$0{\cdot}975 \times 10^{-5}$	
Porosity $7\frac{1}{2}$%	RT 335/00		RT 153A/00		RT 233/00	
$\psi(\xi) = 1$	$1{\cdot}14 \times 10^{-4}$	$0{\cdot}65$–$0{\cdot}9 \times 10^{-4}$	$0{\cdot}64 \times 10^{-4}$	$0{\cdot}4$–$0{\cdot}7 \times 10^{-4}$	$0{\cdot}44 \times 10^{-4}$	$0{\cdot}42$–$0{\cdot}45 \times 10^{-4}$
$\psi(\xi) = 1{\cdot}68$	$0{\cdot}89 \times 10^{-4}$		$0{\cdot}47 \times 10^{-4}$		$0{\cdot}35 \times 10^{-4}$	

TABLE 6

Comparison of calculated and measured nominal values of the specific acoustic resistance ratio for the two perforate porosities: $\psi' = 1$ no interaction effects; $\psi' \neq 1$ interaction included

(N.B.: see Table 1 for comprehensive description of samples)

	Sample description																	
Porosity 22%	RT 335						RT 153A						RT 233					
	Calculated			Measured			Calculated			Measured			Calculated			Measured		
Cavity depth (in)	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3
$\psi'(\xi) = 1$	0·03	0·021	0·017	0·03	0·035	0·035	0·035	0·025	0·02	0·045	0·035	0·038	0·043	0·029	0·025	0·05	0·05	0·045
$\psi'(\xi) = 2{\cdot}95$	0·013	0·009	0·007				0·018	0·012	0·01				0·026	0·018	0·015			
Porosity $7\frac{1}{2}$%	RT 335/00						RT 153A/00						RT 233/00					
$\psi'(\xi) = 1$	0·089	0·062	0·051	0·15	0·1	0·08	0·105	0·074	0·06	0·08	0·07	0·06	0·13	0·091	0·072	0·11	0·13	0·1
$\psi'(\xi) = 1{\cdot}68$	0·058	0·041	0·034				0·073	0·052	0·042				0·098	0·069	0·056			

effect of measured velocity through the hole tends to reduce the effective mass reactance; this trend is also illustrated in Figures 7 to 12.

The results presented in Figures 19 to 24 show the influence of hole diameter for perforates of the same porosity and at the same frequency; these clearly indicate that P_{INC} is not a satisfactory data-collapsing parameter.

One final but extremely important aspect of the work which requires investigation is the estimation of the probable errors in the measured values of the impedance. The theoretical aspects of this work were dealt with in reference [15]. It was noted there that an absolute error of $\pm 0{\cdot}03$ cm was possible in locating the positions of the minima. However, in practice, difficulty in locating the absolute positions of the minima could result in errors in the region of 0·05 cm. Therefore, in assessing the overall error, a much more representative figure for a maximum possible error in minima locations is 0·1 cm. With this figure and representative values for the locations of the minima at the three test frequencies (the distances between minima for the same test frequency for different samples do not change significantly—at least as far as this exercise is concerned), the resulting errors in the real and imaginary impedance can be estimated. As was to be expected the 1 inch cavity measurements are most prone to error, with possible errors of $\pm 5\%$ in resistance and $\pm 10\%$ in reactance. The 3 inch cavity results have possible errors in resistance and reactance of some $\pm 1\%$ whereas the 2 inch cavity results have possible errors in resistance of $\pm 2\%$ and $\pm 2{\cdot}5\%$ in reactance. These error analyses clearly illustrate the reason for scatter in the high frequency (or 1 inch cavity) results and illustrate that these results must be viewed with care when they are used in any detailed assessment.

3.5. INSERTION LOSS MEASUREMENTS

3.5.1. *Introduction*

The original object of these tests, as noted earlier in the text, was to establish or substantiate, if possible, flanking effects through the lightweight honeycomb which is generally used to back perforates. Unfortunately, providing a massive honeycomb (in order to ensure flanking did not occur) resulted in a substantial change in the backing volume to the perforate (compared with the lightweight honeycomb), and this in itself (*via* a change in wall impedance) is sufficient to change the attenuation of the configuration. However, as a result both of carrying out the measurements and of the work involved in calculating the anticipated attenuation, certain interesting aspects have emerged which make the work worthy of inclusion.

3.5.2. *Experimental configuration*

The experimental work was carried out in the impedance tube with the samples under test mounted in the side walls of the tube. The termination of the tube was fitted with an adaptor which held an anechoic termination. This anechoic termination consisted of a wedge of open-cell polyurethane foam. The development of the anechoic termination was carried out experimentally; the geometry of the wedge of foam was modified until microphone traverses, at the frequencies of interest, in the hard-walled tube, showed no trace of a standing wave pattern (i.e., an SWR less than $\frac{1}{2}$ dB).

The arrangement of the samples in the tube is shown in Figure 25. The sample of perforate sits on a 0·030 inch step in the side wall; behind this is placed the honeycomb which is backed by a rigid piston. The surfaces of the honeycomb in contact with the perforate and the backing piston were dipped in a butyl rubber solution before assembly. This ensured that the joints were airtight; it did not interfere with the geometry of the system and did not provide a permanent bond, thus enabling the sample to be used with alternative honeycombs.

The attenuation of the configuration under test was determined by automatically traversing the probe microphone the whole length of the facility, the output being displayed on the level recorder. Thus the spatial variation of the pressure field in the evanescent, test and termination sections could be rapidly assessed.

The honeycombs used in the test were (i) a standard aluminium one with $\frac{3}{8}$ inch cells and 0·004 inch wall thickness, and (ii) a specially manufactured one of $\frac{1}{16}$ inch wall thickness and approximately $\frac{3}{8}$ inch cell size, the complete honeycomb being cast from a lead-based alloy, and therefore having a very high density compared with the aluminium honeycomb. It is worth noting in passing that whilst in principle the latter honeycomb was easy to manufacture, in practice it proved very difficult and many attempts were necessary to achieve the desired result.

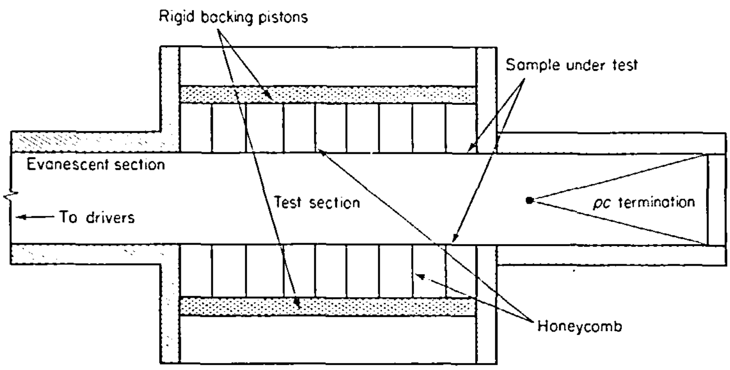

Figure 25. Schematic arrangement of samples mounted in tube for attenuation determination.

3.5.3. *Experimental programme; results*

Two perforates were used in the tests, one of 22% and the other of $7\frac{1}{2}$%. The samples were RT 233 and RT 233/00 (see Table 1), respectively; the material thickness was 0·022 inch. Measurements were carried out with both one and two sides of the tube lined, and with either the lightweight or the massive honeycomb in the backing cavity. The backing cavity depth was 2 inches ($\lambda/4 \simeq 1700$ Hz) and the treated length of section 10 inches (five duct widths). For each configuration the attenuation was measured for five different sound pressure levels at the inlet to the test section (actually measured in the evanescent section).

A typical example of the output from the level recorder is given in Figure 26. This example was for one side lined with RT 233 with the massive honeycomb in the backing cavity. It clearly illustrates the spatial dependence of the pressure field in the three sections: the standing wave in the evanescent section, the attenuation in the treated section—together with the superimposed "ripple", and the nominal standing wave ($\frac{1}{2}$ dB) in the anechoic termination.

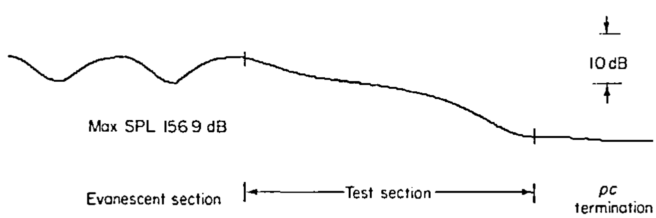

Figure 26. An example of an automatically recorded attenuation. Sample, RT 233; one side lined; massive honeycomb in backing cavity.

The results from the tests are presented graphically in Figures 27 and 28. Results for measured impedance of samples backed by the two different honeycombs are given in reference [16].

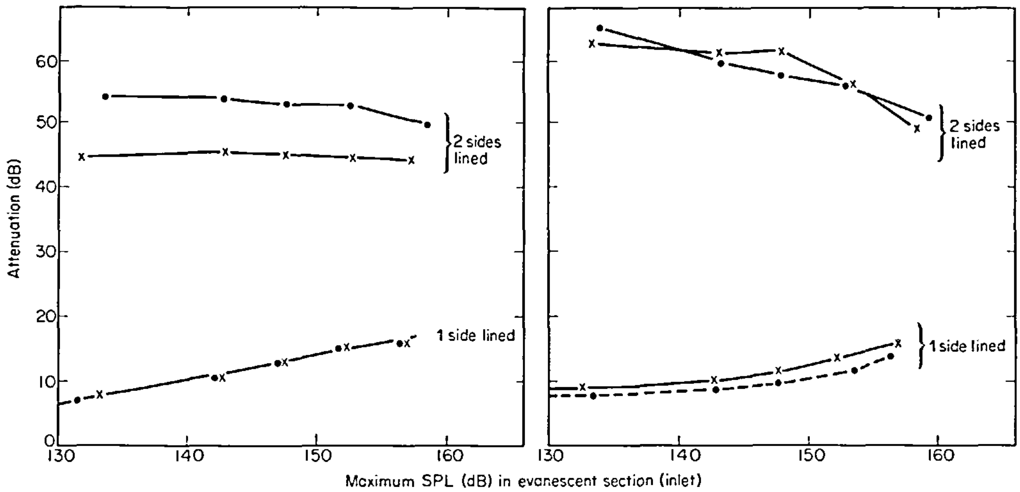

Figure 27. RT 233/00, 7½% porosity. Figure 28. RT 233, 22% porosity.

Figures 27 and 28. Measured attenuations against inlet sound pressure level. ●, Lightweight honeycomb; ×, massive honeycomb.

3.5.4. *Discussion of results*

The interpretation of the data from these measurements is involved and only a semi-quantitative approach is feasible because of some of the theoretical difficulties. Because of the difficulties and limited validity of the theoretical model† for computing the anticipated attenuation rates, the results for only one of the samples has been studied in any detail. This work was carried out for RT 233/00 (see Table 1). Any attempt to interpret the effect of flanking transmission is difficult on account of the modification to the wall impedance by the massive honeycomb resulting in modified attenuation rates.

Several difficulties are immediately apparent in attempting to calculate the attenuation rate. Firstly, if it is assumed that there is no decay of sound pressure level down the attenuating duct, the difficulty is to establish the impedance at the duct wall rather than to attempt to deduce it from some measured value which is dependent on the "free-space" pressure. However, in using the pressure measured at the centre of the duct as a means of establishing the material impedance, with reflection from the termination taken into account, the maximum error would be ±6 dB between it and the "free-space" pressure, with the resultant error in the interpolated impedance. Secondly, with attenuation occurring in the duct the pressure is varying with distance down the duct; so, too, is the wall impedance and hence the attenuation rate. Therefore, in establishing a computer model it is necessary to correct for this by either recomputing at certain incremental steps or, say, assuming that the local impedance in the duct is related to the local mean square pressure gradient. Finally, because the computer programme is for an effectively infinite duct, whereas in practice reflections may occur at the transition from the treated section of the anechoic termination, a possible source of error may arise here. This effect is clearly demonstrated in Figure 26 where standing waves are exhibited in the evanescent section, presumably as a result of the impedance change

† The author wishes to acknowledge the assistance of B. J. Tester in carrying out the computational work.

at the transition between the evanescent section and the treated length of the duct. Similarly the "ripple" superimposed on the decay in the treated section is due to a similar effect at the termination transition (note also that it is unlikely that this is due to the first higher order mode being excited because the microphone, being located centrally, would not tend to respond to this mode).

Because of the aforementioned difficulties it was concluded that as a first attempt the attenuation should be calculated for a case where the material impedance is independent of sound pressure level. If this condition is not upheld at least initially, then, due to the high decay rates in the duct, it is very quickly met. This calculation was carried out for the sample RT 233/00 with two sides lined and for both the massive and lightweight honeycomb. For the light weight honeycomb the measured attenuation was 44·5 dB and the calculated 51 dB and for the massive honeycomb the measured attenuation was 54 dB and the calculated was 60·5 dB. The difference in both cases between measured and calculated values is only about 1·7 dB/duct width and that between the measured and calculated results for the two samples is about 7 dB. Therefore, on the basis of these results, it might be suggested that flanking is not evident. Whilst this is probably as detailed an examination as possible with the present data, nevertheless, in attempting to establish flanking transmission, it was also considered to be of some general interest to establish, at least qualitatively, if the variation in impedance along the treated duct length did result in varying attenuation along the duct length, as indicated by the sound pressure level print-outs.

In order to determine the impedance it was assumed that $SPL_{INC} = SPL_{INLET}$ and that this remained constant down the duct. (These results are presented in Table 4·9 of reference [16].) Further, an estimation was carried out of what length of duct would be traversed by the sound wave at levels below those where the material impedance behaved in a non-linear fashion. (The results are presented in Table 4.10 of reference [16].) From these results it could be seen that the theoretical model, above certain sound pressure levels, begins to underestimate the attenuation. Further, since the attenuation rates at low sound pressure levels are greater than at high sound pressure levels, this trend might be anticipated; indeed, examination of the results shows these trends are consistent since, even at the higher sound pressure levels, some of the duct length (about two and a half duct widths) is traversed by a reduced amplitude wave, for which the impedance is the nominal value and the attenuation rate is much higher. Therefore, the underestimate is worse for higher sound pressure levels than for values nearer the levels where the material is exhibiting its linear impedance everywhere. The higher attenuation rates here are presumably a result of this impedance being nearest the value for optimum attenuation.

It must be emphasized that the above discussion is by no means offered as a comprehensive explanation of this ability of a non-linear liner impedance to give attenuation per unit width that is a function of distance along the duct, nor indeed can the results be taken as a firm indication that flanking is insignificant. It does, however, point to some interesting problems which require solution and may provide some insight into the type of data required for accurate computational work on attenuation rates where the impedance is sound pressure level sensitive.

4. PREDICTION OF THE NON-LINEAR REGIME BEHAVIOUR AND COMPARISON WITH EXPERIMENTAL RESULTS

4.1. HISTORICAL INTRODUCTION

One of the earliest experimental and theoretical accounts of the non-linear behaviour of orifices (in the audio frequency range) was presented in 1935 by Sivian [10]; further contributions were made by Bolt *et al.* [24] in 1949, Ingard and Labate [11] and Ingard [20] in 1950 and 1953, Wood [25] in 1954 and Bies and Wilson [21] in 1957. Significantly, more interest in the subject was shown in the 1960's as a result of the application of Helmholtz resonator arrays to control rocket engine combustion instabilities and the application of perforates in aeroengine intakes for noise control.

The most significant contributions during this time, and up to the present date, were, in chronological order, those of Blackman [26], Ingard and Ising [12], Zorumski and Parrott [27], Ingard [28], Sirignano† [29] and Zinn [30]. Of these, the last two studies [29, 30] were purely theoretical. The earlier works (with the exception of that of Zorumski and Parrott [27] and perhaps that of Ingard [20]) were either experimental or only included "intuitive" attempts at a theoretical understanding, usually stemming from the ideas put forward by Sivian [10].

In his analysis, Sivian, on the basis of the velocity dependence of the resistance as observed from his experimental results, assumed that the relative loss in kinetic energy of the flow in the orifice could be equated to the resistance of the orifice. Sivian then demonstrated that, for the isentropic flow case, it is a simple matter to show that the additional resistance, due to this loss of kinetic energy, is $\rho u/2$, where u is the mean particle velocity in the hole. Comparing the measured and calculated orifice resistance (based on this additional resistive term) Sivian concluded that "the non-linear behaviour of the resistive component of the impedance can be partly accounted for on the basis of kinetic energy of flow. Turbulence may also play a part, but it need not be the sole cause" [10].

It is not apparent what Sivian precisely means by the phrase "turbulence may also play a part". Bolt *et al.* [24] throw no further light on this question but simply appear to accept Sivian's interpretation of the situation.

The work reported by Ingard and Labate [11] is a detailed experimental study of the flow field in and around the orifice. They presented some of their results in the form of "phase diagrams" showing the existence of several possible flow regimes for given particle velocities. This work was significant because it presented photographic evidence of the flow field of the orifice, which showed vortex ring generation and formation of a "jet". This quantitative analysis of the orifice losses, and the consequent increases in resistance, were based on equating the sum of the translational energy in the jet and that of the rotational energy in the vortex rings to an effective acoustic energy loss for the orifice. Thus, this acoustic energy loss was dependent on the mean square acoustic particle velocity in the orifice. In their actual calculations, the rotational energy term was neglected. The translational energy term was experimentally determined by means of a torsion disc measurement of the jet momentum and an indirect stroboscopic measurement of the jet velocity. Their comparison of experimental and theoretical results showed "rather good agreement" [11]. However, they did not attempt to explain, in any detail, the non-linear behaviour in terms of more fundamental parameters such as, for example, the mean acoustic particle velocity in the orifice.

Ingard's 1953 paper [20] describes a major study of resonators. One small part of this study was concerned with the non-linear behaviour. The interpretation given of this is based on the experimental work reported in the previous paper [11]. The quantification of this work was improved, however, by basing it on the acoustic particle velocity in the aperture (orifice). In terms of this parameter, Ingard [20] found that the non-linear resistance is of the form $R_s[0{\cdot}7(u/100)^{1{\cdot}7}]$, where R_s is the usual Helmholtz viscous term $(2\mu\rho\omega)^{1/2}$, u being the acoustic particle velocity amplitude in the aperture. Apart from the fact that, according to this formula, the non-linear resistance is now dependent on frequency, as distinct from Sivian's analysis, the velocity dependence is also significantly increased, from an index of 1 to 1·7. Ingard also commented on the observed trend of decreasing mass reactance above certain

† Sirignano's work was actually carried out in 1966. However, it was not reported in the general technical literature and therefore remained generally unknown. The order in which it is presented here perhaps reflects more accurately its appreciation in the scientific world in relation to other studies.

sound pressure levels, but did not attempt any comprehensive quantitative interpretation of this.

The work of Wood [25], whilst carried out for steady flow only, is of interest because of some of the conclusions that he reaches. In his measurements on orifices he found that, in all cases, a transition from a velocity independent resistance to a velocity dependent resistance occurred at a Reynolds number (based on orifice diameter) of about 10, a result which should carry over to the alternating flow case if a quasi-steady flow situation can be assumed. He also found that the velocity independent resistance for steady flow could be explained in terms of the usual acoustic end corrections: i.e., the effective length of the orifice should be taken to be equal to its physical length plus the acoustic end correction.

Bies and Wilson [21] investigated the non-linear behaviour of a resonator mounted in an impedance tube in two different configurations: (a) at the end of the tube and (b) in the side of the tube. It was found that when the resonator was mounted in the side of the tube the non-linear resistance term agreed very well with that suggested by Sivian. With the resonator mounted in the end of the tube, however, the resistance was found to be higher; this is perhaps not surprising in view of the fact that in the case of the side mounted resonator the confining wall of the tube may have interfered with the orifice jet. They also found that the mass reactance was sensitive to acoustic particle velocity, tending to decrease indefinitely with increasing velocity, above a certain velocity. Also, this velocity sensitivity of the reactance depended on the orientation of the resonator (the side wall mounted condition being more velocity sensitive). No explanation of this sensitivity of mass reactance to velocity was given.

Blackman [26] carried out measurements on perforated plates and attempted to use Ingard's theory [20] to explain the behaviour. The agreement he obtained was very poor. Blackman suggested the following reason for this bad agreement: "Presumably this results from the coupling of the multiple resonators". Blackman concluded that the velocity dependence had an index of 0·4 (as against Ingard's value of 1·7). Further, in his investigation of mass reactance, he found that the best correlation parameter was percentage open area, rather than sound pressure level or particle velocity. Curiously he appears to have interpreted Ingard's empirical non-linear correction factor for the mass reactance as an end-effect correction factor!

In the work reported in their joint paper, Ingard and Ising [12] not only examined the effect of high sound pressure levels on the orifice resistance but also that of a biasing steady flow; also a hot-wire anemometer was used to measure directly the acoustic particle velocity in the aperture. They concluded that the expression for the non-linear resistance given by Ingard in a previous paper [20] is valid provided that the contribution to the resistance by the viscous term is small. For high aperture particle velocities, however, they found that the formula given by Sivian [10] is a better approximation. In connection with the mass reactance behaviour, they comment as follows: "At high velocities, which according to our experimental results means $u_0 > 10$ m/s (note that here u_0 is the peak particle velocity), the mass reactance decreases and approaches a value that we expect to be approximately one-half the linear value. This expectation is based on the assumption that, in the high velocity region, the flow at any moment is irrotational only on the side of the orifice where at the moment the flow is towards the orifice. Thus associating the acoustic inertia of the flow with the irrotational portion leads to the reduction factor of approximately one-half" [12].

(Chronologically, the work of Zorumski and Parrott [27] should be considered next, but it is preferable to describe this following discussion of the theories of Sirignano [29] and Zinn [30], to be presented later.)

In a further study of the question [28], Ingard, without presenting any justification for a quasi-steady analysis, formulates (although not explicitly) the non-linear resistance in a form similar to that presented by Sivian [10], but modified for non-ideal flow by means of an orifice discharge coefficient. The resultant non-linear resistance term at the fundamental frequency, which may be extracted from Ingard's work [28], is $(1{\cdot}11/C_D^2)(\rho/2)u_{max}$, where U_{max} is the acoustic particle velocity amplitude of the flow in the orifice and C_D is the "orifice coefficient" [28]. It is perhaps interesting to note this rather casual use of the discharge coefficient and its similarly imprecise definition, which could be somewhat misleading; precise relevant definitions can be given and they will be presented later in this chapter.

Two of the three most important recent theoretical studies, those of Sirignano [29] and Zinn [30] (that of Zorumski and Parrott [27] being the third), are very similar; in both cases it is assumed that there is an additional dissipative term due to the "jet" flow field in the region of the orifice. Whilst the respective non-linear resistances predicted by the analyses differ by precisely a factor of two (this will be commented on shortly), it is important to note that in both cases the assumption of ideal jet flows is made: i.e., no explicit account is taken of turbulence and viscosity is included only by way of the usual, Kirchoff–Helmholtz viscous shear in the orifice itself. For example, according to the models used by Zinn and Sirignano, the flow through the orifice in the steady flow case would not display all the losses implied by the actual discharge coefficient. The two theories use essentially the same methods, and, for the most part, are internally self-consistent and logical deductions from the respective assumptions made. Sirignano's theory appears completely satisfactory in this respect. Zinn, in obtaining his second-order momentum equation appears to use an unstated assumption to the effect that the orifice is infinitely thin.

The essential differences between the theories can be traced to the respective explicit and implicit assumptions made. Sirignano assumes (in effect explicitly) that the second-order pressure changes associated with the flows both into and out of the ends of the orifice incorporate a momentum flux term of Bernouilli type: that is, the total Bernouilli pressure, $p + \rho u^2/2$, is conserved to second order along streamlines. Zinn's assumptions, on the other hand, appear to lead, in an implicit and rather obscure fashion, to the result that, on the inflow side of the orifice, the second-order pressure change is twice that used by Sirignano. In the final result, this leads simply to Zinn's prediction of the second-order non-linear resistance being exactly twice Sirignano's.

The approach of Zorumski and Parrott [27] is completely different to those of Zinn and Sirignano, in that the measured steady pressure loss across the sample (together with the nominal impedance value) is used to determine the acoustic non-linear impedance. The analysis is based on the assumption that what they call the "instantaneous acoustic laws" for velocity and pressure must be obeyed across the sample and that on either side of the sample the velocity potential must satisfy the time dependent wave equation. These "instantaneous acoustic laws" imply incompressibility insofar as conservation of mass across the sample is concerned and include a semi-empirical term in the momentum conservation expression. By writing the velocity and pressure at the sample in terms of a Fourier series to include all harmonics of the signal, and by carrying out appropriate iterative processes, Zorumski and Parrott then obtain pressure and velocity coefficients as functions of the fundamental driving frequency: i.e., the non-linear impedance at each driving frequency is obtained, with the non-linear terms being explicitly associated with the amplitude distribution of the higher harmonics generated.

4.2. THEORETICAL ASPECTS

4.2.1. *Introduction*

It is evident from the review presented in section 4.1 that the work on non-linear acoustic behaviour has, primarily, been concerned with a single isolated and, in the majority of cases, ideal orifice. Further, the theoretical modelling can, broadly speaking, be divided into two approaches; these might be loosely described as (a) simple energy conservation modelling and (b) sophisticated mass, momentum and energy conservation modelling.

Typical examples of the energy based approach (a) are those of Sivian [10] and Ingard [11]. The models based on this approach yield no details as to the mechanisms involved in the dissipation of energy. In Sivian's cases it is explicitly assumed that all of the kinetic energy based on the orifice mean acoustic velocity is dissipated, and it is implicitly assumed that the orifice losses are those appropriate to a low Reynolds number. It is therefore inevitable that for a given incident pressure field the controlling parameter in determining the non-linear resistance for a perforated sheet is its porosity.

The more sophisticated theories (b), for example those of Zinn [30] and Sirignano [29], are precisely defined and internally logical; the weaknesses in these cases ultimately arise from the assumptions, both explicit and implicit, about the flow field external to the orifices. Specifically, these include the neglect of viscosity and turbulence, and consequential errors necessarily resulting from the departures from ideal jet flow conditions in the true physical situation. This can be illustrated more clearly if, for the present, it can be assumed that the flow in the orifice can be taken to be quasi-steady (this will be commented on in more detail later in this section). Under these conditions (and provided the orifice thickness is small compared with its diameter) the flow field can be visualized as corresponding to a steady flow field, which is a case which is relatively well understood [31]. It is found in practice that for Reynolds numbers in excess of approximately 10 the loss in the Bernouilli constant across a sharp edged orifice plate differs from that which might be anticipated from simple conservation of momentum and energy in an ideal fluid (Bernouilli's theorem). Batchelor [32] has carried out such an analysis for the ideal case and has shown that the loss through a sudden transition (i.e., orifice) as compared with a gradual transition (i.e., lossless system where Bernouilli's theorem applies) is proportional to $\rho u^2/2$, where u is the mean upstream (and downstream) velocity far from the orifice. The exact meaning of the term "ideal flow" implied, for example, in Batchelor's case, is simply an assumption which is indirectly related to viscosity effects but specifically is concerned with the radial velocity component of the flow through the orifice. The flow into the orifice from the upstream end will have a radial component, which will be dependent on the ratio of the orifice area to the effective free stream area; this in general will result in the flow either into the orifice (if it is long) or leaving the orifice also having a radial component. This radial component, which is towards the axis of the orifice, ultimately causes a contraction of the flow (or jet) and eventually a minimum flow area is formed where the streamlines are parallel; this minimum area is called the *vena contracta*. At this point the flow velocity is at a maximum whilst the "active" flow area is at a minimum. The ratio of the area of the *vena contracta* to the orifice area is known as the coefficient of contraction, C_a, for the orifice [33]. Its value, as might be anticipated, is dependent on the static head across the orifice, on the ratio of orifice area to free stream area and on the fluid viscosity. In practice it is also found that the velocity in the *vena contracta* is less than that which might be anticipated from the assumption of an ideal inviscid fluid. This ratio of the ideal velocity to actual velocity at the *vena contracta* is known [33] as the coefficient of velocity, C_v. Whilst the coefficient of velocity is a function of pressure ratio it is generally very nearly equal to unity. In practice, because of the standard uses of orifices for measuring flow rates, a third coefficient is generally exclusively quoted; this is known as the discharge coefficient, C_D, and it is equal to the product of the velocity and contraction coefficients.

It is reasonable to assume that these considerations about mean flows may be carried over to the acoustic case. For example, examination of the photographs of the orifice acoustic flow fields in reference [11] appear to indicate the formation of *vena contractas*. On this basis of argument, the limitations of the theories of Sirignano, Zinn and Zorumski follow directly, although the detailed way in which the real behaviour may be in conflict with their assumptions naturally depends on the particular types of modelling used by these authors.

In the light of all these past results, it is possible, and indeed appears reasonable, to proceed theoretically on the following basis. First, major weight in establishing physically valid parameters for the non-linear impedance should be given to the results from the well-understood cases of steady flow through orifices. Second, any theoretical formulas to be derived must contain the full, correct linearized results (including viscous effects) of Helmholtz, Kirchoff and Rayleigh. Third, orifice interactions should contain the Fok-type interaction term in the linearized limit. Finally, the theoretical formula should be in terms of accurately measurable parameters.

4.2.2. *Parametric dependence of the non-linear behaviour*

Quite obviously, any theoretical derivation of the non-linear impedance (or in general any relevant parametric dependence) of materials will only include those effects implicitly accounted for and included in the initial equations. Specifically, the initial assumptions and approximations ultimately determine the form, accuracy and detail of the final result. It is therefore desirable, when establishing the initial equations to examine in some detail the assumptions and approximations being made.

In the light of the requirements listed in the previous section for establishing a model for the non-linear behaviour of the impedance it is convenient, as an initial step, to derive a suitable form of the linear momentum balance equation for the fluid. From this equation, subject to the explicitly stated assumptions, a form for the non-linear impedance of the material may be deduced.

For a Stokesian fluid, in the absence of external forces, the exact linear momentum balance equation, in the fluid, may be written in vector form as follows:

$$\rho\frac{\mathrm{D}v_i}{\mathrm{D}t}+\frac{\partial p}{\partial x_i}-\frac{\partial}{\partial x_j}\left[\mu\left(\frac{\partial v_i}{\partial x_j}+\frac{\partial v_j}{\partial x_i}-2\frac{\partial v_k}{\partial x_k}\delta_{ij}\right)+(\tfrac{4}{3}\mu+\zeta)\frac{\partial v_k}{\partial x_k}\delta_{ij}\right]=0. \tag{19}$$

Here the terms have their usual notation, μ† and ζ being the shear and bulk viscosities, respectively. The above equation is, of course, the Navier–Stokes equation.

If μ is effectively constant in time (i.e., slowly varying or actually constant), then μ can be replaced by $\bar{\mu}$. Further, by applying the conservation of mass equation,

$$\frac{1}{\rho}\frac{\mathrm{D}\rho}{\mathrm{D}t}+\frac{\partial v_i}{\partial x_i}=0 \tag{20}$$

the equation (19) may be written as

$$\rho\frac{\mathrm{D}v_i}{\mathrm{D}t}+\frac{\partial p}{\partial x_j}+\bar{\mu}(\nabla\times\nabla\times\mathbf{v})_i+\frac{\partial}{\partial x_i}\left[(\tfrac{4}{3}\mu+\zeta)\frac{1}{\rho}\frac{\mathrm{D}\rho}{\mathrm{D}t}\right]\doteq 0. \tag{21}$$

From thermodynamics, it can be assumed that, for fluids, ρ can be expressed as a function of the pressure, p, and entropy, S:

$$\rho=\rho(p,S). \tag{22}$$

Consequently, if entropy variations are negligible,

$$\frac{1}{\rho}\frac{\mathrm{D}\rho}{\mathrm{D}t}\doteq\frac{1}{\gamma}\frac{1}{p}\frac{\mathrm{D}p}{\mathrm{D}t}, \tag{23}$$

† Note that in section 2 the symbol μ was used to indicate an effective viscosity coefficient (not necessarily the shear viscosity).

where γ is the ratio of specific heats. Hence equation (21) may be written as

$$\rho\frac{\mathrm{D}v_i}{\mathrm{D}t}+\frac{\partial p}{\partial x_i}+\bar{\mu}(\nabla\times\nabla\times\mathbf{v})_i+\frac{\partial}{\partial x_i}\left[\frac{(\frac{4}{3}\mu+\zeta)}{\rho}\frac{\rho}{\gamma p}\frac{\mathrm{D}p}{\mathrm{D}t}\right]\doteq 0. \quad (24)$$

The last term on the left-hand side of equation (24) may be taken to be negligibly small. It represents the diffusion of acoustic momentum in a viscous medium giving rise to negligibly small attenuation of the acoustic waves.

Therefore, a good and sufficient approximation to the linear momentum balance in the region of the orifices and within the orifices may be written as

$$\rho\frac{\mathrm{D}v_i}{\mathrm{D}t}+\frac{\partial p}{\partial x_i}-\bar{\mu}\nabla^2 v_{i_{\text{solenoidal}}}\doteq 0. \quad (25)$$

Equation (25) represents an initial step in the analysis; as noted in the previous section any interpretation of the non-linear impedance should be in terms of well-established parameters and also easily determined ones. On the basis of the forthcoming results (and on the basis, to some extent of the previous analyses), ideally the interpretation should lead ultimately to the non-linear impedance (at least for the resistive part) in terms of the mean velocity in the hole or aperture. Thus ideally it is required that:

$$v_i=[v_r=0,\ v_\theta=0,\ v_x(r,x,t)], \quad (26)$$

where ultimately $v_x(r,x,t)$ is replaced by some velocity which is suitably spatially averaged.

It would, of course, be more accurate to use the definition of equation (26) only for the region confined to the hole: i.e., for regions external to the hole, v_r should, perhaps, be allowed to take on some finite value. Since, however, it is known that end-effects are concerned with the fluid mass only within approximately one hole radius distance of the hole, neglecting v_r outside the hole should not give rise to large errors. In practice, errors should be small if the interpretation is carried out with an end-effect approximation: i.e., "an equivalent hole extension" which compensates for the radial velocity component (e.g., in the linear regime, as noted by Sivian [10], this equivalent hole extension or end-effect region adequately compensated for the radial velocity component).

With this assumption, and a further assumption that the density in equation (25) can be approximated by the mean density (i.e., $\rho=\bar{\rho}+\rho'\doteq\bar{\rho}$), then equation (25) with equation (26) becomes

$$\bar{\rho}\frac{\partial v_x(r,x,t)}{\partial t}+\bar{\rho}v_x(r,x,t)\frac{\partial v_x(r,x,t)}{\partial x}+\frac{\partial p}{\partial x}-\bar{\mu}\nabla^2_{r,x}(r,x,t)\doteq 0. \quad (27)$$

In order to rewrite equation (27) in terms of the appropriate space averaged quantities the identity

$$\langle f\rangle_A(x,t)\equiv\frac{1}{A}\int_A f(r,\theta,x,t)\,\mathrm{d}A$$

will be employed. Then, rewriting equation (27) in terms of cylindrical or polar coordinates and carrying out the appropriate integration yields

$$A\bar{\rho}\frac{\partial}{\partial t}[\langle v_x\rangle_A(x,t)]+A\bar{\rho}\frac{\partial}{\partial x}[\langle\tfrac{1}{2}v_x^2\rangle_A(x,t)]+A\frac{\partial}{\partial x}[\langle p\rangle_A(x,t)]-$$

$$-A\bar{\mu}\left\{\frac{1}{A}2\pi\left(r\frac{\partial v_x(r,x,t)}{\partial r}\right)\Bigg|_{r=0}^{r=R}+\frac{\partial^2}{\partial x^2}[\langle v_x\rangle_A(x,t)]\right\}=0. \quad (28)$$

Note that for the purpose of this exercise the hole has been taken to have a radius R and area A. Define

$$\langle v_x\rangle_A(x,t)\equiv v(x,t),\quad \langle p\rangle_A(x,t)\equiv p(x,t). \tag{29}$$

By dividing equation (28) by A and using equation (29), equation (28) can be written as

$$\bar{\rho}\frac{\partial v(x,t)}{\partial t}+\bar{\rho}\frac{\partial}{\partial x}\{[\tfrac{1}{2}v^2(x,t)]+[\langle\tfrac{1}{2}v_x^2\rangle_A-\tfrac{1}{2}v^2(x,t)]\}+\frac{\partial p(x,t)}{\partial x}-$$
$$-\bar{\mu}\left[\frac{2\pi R}{A}\left(\frac{\partial v_x}{\partial r}\right)(R,x,t)+\frac{\partial^2 v(x,t)}{\partial x^2}\right]=0. \tag{30}$$

Assume

$$\langle\tfrac{1}{2}v_x^2\rangle_A \doteq \tfrac{1}{2}v^2(x,t): \tag{31}$$

i.e., the square of the average velocity over the hole cross-section is approximately equal to the square of the velocity averaged over the hole cross-section. This will be true if the velocity profile is flat and falls sharply near the walls (i.e., "plug" flow) but not necessarily true for parabolic flow profiles (i.e., Poiseuille flow). However, for the frequencies of concern here, the boundary layer thickness is small (see section 2) compared with the hole radius. Hence equation (31) is sufficiently accurate. Note, however, that this is not necessarily so for materials like "Rigimesh" (porous fibre type) having small pore size, and consequently a small equivalent hole diameter. The last term on the left-hand side of equation (30) can be seen, by inspection, to contain viscous terms both of the Poiseuille and Helmholtz types, respectively (see, for example, reference [34]). This term, however, for the general case will have the form derived by Crandall [19], the two forms of resistance noted above being the limiting cases of Crandall's formula. It will therefore be assumed that this term may be replaced by a general, velocity independent, specific resistance term. The choice, or substitution, for this term, will depend on the particular circumstances. Consequently

$$\bar{\mu}\left[\frac{2\pi R}{A}\left(\frac{\partial v_x}{\partial r}\right)(R,x,t)+\frac{\partial^2 v(x,t)}{\partial x^2}\right]\equiv R_\mu v. \tag{32}$$

R_μ in practice may, of course, be most suitably chosen from experimental values; however, it is specifically associated with the linear regime resistance.

Finally, with $v\equiv v(x,t)$, equation (30) becomes

$$\bar{\rho}\frac{\partial v}{\partial t}+\bar{\rho}\frac{\partial}{\partial x}(\tfrac{1}{2}v^2)+\frac{\partial p}{\partial x}+R_\mu v=0. \tag{33}$$

Equation (33) represents the linear momentum balance in the region of the hole and is "exact" within the limits of the explicitly stated approximations. Indeed the equation is exactly what might have been anticipated save for the second term on the left-hand side which represents possible additional losses resulting from the kinetic energy of the flow. It is from this term that the ideal losses derived by Sirignano and Zinn were obtained.

Before proceeding further with the analysis it is therefore desirable to modify equation (33) to take into account real physical effects (i.e., known radial velocity effects) and to state explicitly the conditions under which the modifications are valid.

If v is assumed to have simple harmonic time dependence and the gradients are assumed to be effectively constant ($\partial/\partial x \to 1/l$ where l is some length scale: i.e., $l \ll \lambda$ which is always true) then equation (33) may be rewritten as

$$[i\omega\bar{\rho}l+R_\mu l+\bar{\rho}v/2]\,v \doteq -p. \tag{34}$$

Physical interpretations for the terms in the bracket on the left-hand side now become apparent as follows. l is taken to represent (at least for the purposes of this analysis) the effective length of orifice (i.e., its physical thickness plus the end effects). The first term in the bracket is therefore the specific acoustic mass reactance. The second term is the specific acoustic resistance and the third term is the contribution to the specific impedance of the kinetic energy loss associated with the flow through the hole. Further, if the flow in the orifice can be taken to be quasi-steady, this latter term can be more specifically identified with the corresponding steady flow losses through an orifice and hence take into account the radial component of the flow. In any case, continuing the analysis based on the present form of equation (34) would yield a result similar to that of Sirignano.

The flow in the hole (orifice) may be considered as quasi-steady if the time derivative term in the Euler equation for perturbed flow (acoustic field) is small in comparison with the spatial derivative terms. Euler's equation may be written as

$$\bar{\rho}\frac{\partial u'}{\partial t} + \bar{\rho}u'\frac{\partial u'}{\partial x} + \frac{\partial p'}{\partial x} = 0; \tag{35}$$

the bar denotes average qualities and the prime the fluctuating components. From order of magnitude considerations the ratio of the first two terms in equation (35) is $(\hat{l}'/u')/T$, where $\hat{l}'$ is a characteristic fluid displacement, u' the amplitude of the velocity perturbation and T the oscillatory period. For the high velocity region this ratio is clearly small even at the high frequencies since the characteristic fluid displacement should ideally be confined to the orifice thickness (i.e., velocities $O(10^3)$ cm/s, $T \sim O(10^{-3})$ s and $\hat{l}' \sim O(10^{-1})$ cm).

Accordingly equation (34) may be rewritten as

$$[i\omega\bar{\rho}l + R_\mu l + K|v|]\,v = -p, \tag{36}$$

where the coefficient $\rho/2$ has been replaced by an effective mass density parameter, K, which will be derived from observations on steady flows through orifices.

The desired form of the non-linear resistance can now be obtained by equating the respective energy flow in the orifice (hole) at some "control" area external to the hole where the classical or linear acoustics laws are valid. This control area does not have to be taken at any particular physical location. The energy flux in the hole is, of course, derived *via* the linear momentum balance equation (36) for the hole.

By referring to Figure 29 and using equation (36) one can write the linear momentum balance for the hole as

$$p_-^{(a)}(t) = \rho c M\frac{\partial v_{\mathrm{H}}(t)}{\partial t} + R_\mu v_{\mathrm{H}}(t) + K|v_{\mathrm{H}}(t)|v_{\mathrm{H}}(t), \tag{37}$$

where M is the specific mass reactance ratio of the hole. For the acoustic zone area external to the hole, where the "classical" acoustic relationships apply

$$p_-^{(a)}(t) = \rho c\left[R + M\frac{\mathrm{d}}{\mathrm{d}t}\right]v_n^{(a)}(t), \tag{38}$$

where R and M, respectively, are the specific resistance and reactance ratios of the zone area. Also, from continuity of volume velocity (see Figure 29)

$$S_C v_n(t) = S_{\mathrm{H}} v_{\mathrm{H}}(t). \tag{39}$$

Now the mean energy flux over the cycle is

$$\frac{1}{T}\int_{t}^{t+T} p(t)\,v(t)\,\mathrm{d}t, \tag{40}$$

where T is the period.

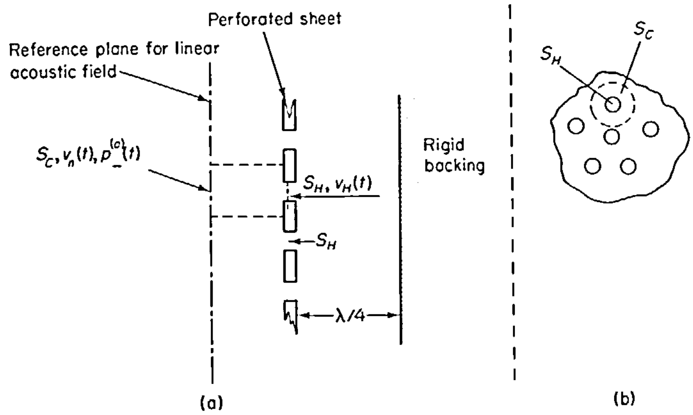

Figure 29. Diagrammatic representation of: (a) perforate and backing cavity used for theoretical model of non-linear resistance; (b) hole zone area. S_H = area of hole; S_C = zone area of hole; v_H = velocity amplitude in hole; v_n = velocity amplitude in zone area; $p_-^{a}(t)$ = pressure amplitude applied to front face of perforate.

From equation (38) and by using equation (40) this becomes, for the zone area,

$$\frac{1}{T}\int_{t}^{t+T} p^{(a)}(t)\,v_n^{(a)}(t)\,\mathrm{d}t = \rho c R[v_n^{(a)}]^2_{\text{r.m.s.}} + 0. \tag{41}$$

In determining the energy flux for the hole it is assumed that the instantaneous non-linear relationship between the pressure and velocity in the hole is of the quasi-steady character as previously described.

From equation (39)

$$S_C\frac{\mathrm{d}v_n}{\mathrm{d}t} = S_H\frac{\mathrm{d}v_H}{\mathrm{d}t}. \tag{42}$$

Consequently, equation (37) can be written solely in terms of $p_-^{(a)}(t)$ and $v_n(t)$. Further, by using equation (40) the energy flux through the hole may be written as

$$\frac{1}{T}\int_{t}^{t+T} \rho c M\left(\frac{S_C}{S_H}\right)\frac{\mathrm{d}v_n(t)}{\mathrm{d}t}v_n(t)\,\mathrm{d}t + \frac{1}{T}\int_{t}^{t+T} R_\mu\left(\frac{S_C}{S_H}\right)v_n(t)\,v_n(t)\,\mathrm{d}t$$
$$+\frac{1}{T}\int_{t}^{t+T} K\left(\frac{S_C}{S_H}\right)^2 |v_n(t)|[v_n(t)]^2\,\mathrm{d}t. \tag{43}$$

If M is linear the first term in expression (43) is zero; hence, expression (43) reduces to

$$R_\mu\frac{V_n^2}{2} + \frac{1}{T}\int_{t}^{t+T} K\left(\frac{S_C}{S_H}\right)^2 V_n^3\sin^3\left(\frac{2\pi t}{T}\right)\mathrm{d}t, \tag{44}$$

where $v_n(t) = V_n \sin(2\pi t/T)$. In rewriting expression (43) in the form (44) it has been assumed that $v_n(t)$ is harmonic. This is reasonable since by definition v_n is defined for the linear acoustic region.

Finally, carrying out the integration of the second term in expression (44) yields

$$R_\mu \frac{V_n^2}{2} + \frac{4}{3\pi}\left[K\left(\frac{S_C}{S_H}\right)^2\right] V_n^3. \tag{45}$$

Equating the results of the integrations of equation (38) and (37) yields

$$\rho c R \frac{V_n^2}{2} = R_\mu \frac{V^2}{2} + \frac{4}{3\pi}\left[K\left(\frac{S_C}{S_H}\right)^2\right] V_n^3, \tag{46}$$

where the r.m.s. velocity in equation (41) has been replaced by the O-peak velocity. Consequently, from equation (46),

$$R = \frac{1}{\rho c}\left\{R_\mu + \frac{8}{3\pi}\left[K\left(\frac{S_C}{S_H}\right)^2\right] V_n\right\}. \tag{47}$$

It is evident from equation (47) that the effective hole specific acoustic resistance ratio has two components: (i) R_μ, which is the classical viscous resistance and (ii) a component which is velocity dependent. It remains now to determine the factor K so that the velocity dependent resistance may be written explicitly in terms of known parameters quantifying the flow behaviour in the hole.

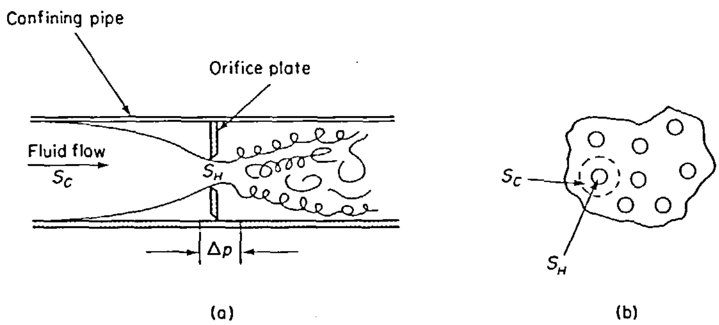

Figure 30. Schematic drawing of: (a) a conventional orifice plate system; (b) the equivalent system for a perforated plate.

The parameters controlling the flow of fluids through orifice plates (that is, through restrictions in pipes) is well reported in the literature [31, 33]. In particular the information is presented for so-called "sharp edge" orifices. For present purposes it is assumed that the orifice is sharp edged; departures from this requirement and the effect on predictions will be dealt with in a subsequent section.

The behaviour of orifice plates has been studied in some detail [31] as a result of their applications to measuring fluid flow rates in piping systems and the like. The system essentially consists of a pipe with a diaphragm across it in which there is a hole. By monitoring the pressure differential across the hole (diaphragm) the fluid flow rate may be determined: i.e., for any given flow condition the ratio of the pressure drop to the velocity is thus determined. Therefore, if the steady state and instantaneous acoustic behaviour can be taken to be equivalent, then the acoustic impedance can be derived from the steady-state behaviour.

The common configuration of an orifice–plate system is shown in Figure 30(a); in Figure 30(b) is shown what is taken, for the present, to be the equivalent system for the perforated plate, where the control or zone area (unit area divided by the number of holes per unit area) is taken to be equivalent to the orifice area. The pressure loss across the orifice can be written [31] as

$$\Delta p = \frac{1}{[C_D(\mathrm{Re}N_0)]^2}\left(\frac{S_C}{S_H}\right)^2\left[1-\left(\frac{S_H}{S_C}\right)^2\right]\rho\frac{\bar{v}_n^2}{2}, \tag{48}$$

where C_D is the discharge coefficient for the orifice (a function of Reynolds number) and $\bar{v}_n$ is the steady-state (d.c.) velocity. The ratio S_H/S_C is the porosity, P, of the perforate. Further, the equivalent mass density parameter K can now be defined for the flow through the orifice. Writing equation (48) in terms of the velocity in the orifice and the porosity gives

$$\Delta p = \frac{1}{C_D^2}\frac{1}{P^2}(1-P^2)\frac{\rho}{2}(P\bar{v}_H)^2. \tag{49}$$

Then, comparing equations (49) and (36) yields

$$K = \frac{1}{C_D^2}(1-P^2)\frac{\rho}{2}. \tag{50}$$

Consequently, equation (47) for the specific acoustic resistance ratio may be written as

$$R = \frac{1}{\rho c}\left\{R_\mu + \frac{8}{3\pi}\left[\frac{1}{C_D^2}\left(\frac{1-P^2}{P^2}\right)\left(\frac{\rho}{2}\right)V_n\right]\right\}. \tag{51}$$

The resistance, for reasons which will be apparent later in the text, has in this instance been written in terms of the root mean square velocity in the hole; consequently it is more convenient to rewrite equation (51) as (also the discharge coefficient is dependent on the Reynolds number based on hole diameter and velocity)

$$R = \frac{1}{\rho c}\left[R_\mu + \frac{1{\cdot}2}{C_D^2}\left(\frac{1-P^2}{P}\right)\frac{\rho}{2}(v_H)_{\mathrm{r.m.s.}}\right]. \tag{52}$$

It is interesting to note that, when the terms resulting from the non-ideal flow case are disregarded (i.e., $(1-P^2)/(C_D^2P^2)$), the non-linear resistance depends on velocity in a way identical to that derived by Sirignano [29]: namely $(4\rho/3\pi)$, which is one-half the value obtained by Zinn [30]. As noted earlier in the text the problems of non-ideal (i.e., non-sharp edged) orifices and of interaction effects will be dealt with in section 4.3.1.

4.2.3. *Parametric dependence of the non-linear reactance*

The development of any formulas for behaviour of the non-linear mass reactance, with, for example, velocity as the dependent variable, is difficult. It has received little attention presumably because for thin plates it is small. In any case—as far as is evident from experimental work—it tends to an asymptotic limiting value of approximately one-half the linear regime value (the value, that is, for a thin plate with a hole diameter large compared with the hole (plate) thickness). Westervelt [35] has estimated quantitatively that "when the particle displacement amplitude through the orifice becomes approximately greater than the diameter of the orifice the coherence of the mass within the hemispherical caps will be destroyed by the jet flow which is known to occur [11]. This will remove 5/8 of the kinetic mass of a thin orifice, and hence the reactance of the orifice will be reduced by about 5/8 of its linear value." In reaching this conclusion Westervelt has approached the problem of the impedance of orifices through energy functions. Through this approach he deduced that the mass reactance is composed of two components: (i) a term due to the kinetic energy in the region external to the attached mass region and (ii) a term due to the attached mass. The respective fractions are 3/8 and 5/8.

Consequently, since the phenomena is based on the onset of turbulence in the jet formed by the orifice it would be reasonable to assume that the behaviour will be Reynolds number dependent. Thus, typically, for Reynolds numbers in the range 10–100, based on hole diameter, [36], this phenomenon of decreasing mass reactance should be observed.

In order to examine the possibility of the present method of analysis yielding any non-linear dependence of the mass reactance it is necessary to return to equation (33): namely,

$$\bar{\rho}\frac{\partial v}{\partial t}+\bar{\rho}\frac{\partial}{\partial x}(\tfrac{1}{2}v^2)+\frac{\partial p}{\partial x}+R_\mu v \doteq 0.$$

Any non-linear behaviour, at least at the fundamental frequency, is clearly only possible as a result of the first term on the left-hand side of this equation, but in its exact form (see equation (25)), with $\bar{\rho}$ replaced by $\rho_0+\rho'$:

$$(\rho_0+\rho')\frac{\partial v}{\partial t}=\rho_0\left(1+\frac{\rho'}{\rho_0}\right)\frac{\partial v}{\partial t}, \tag{53}$$

and, for purposes here, with also

$$v = V_0 \sin \omega t$$

and

$$\frac{\rho'}{\rho_0}=\alpha\sin(\omega t+\beta),$$

where β represents some phase lag (or lead), and α represents the ratio of the steady to the fluctuating density. Hence expression (53) now becomes

$$i\omega\rho[1+\alpha\sin(\omega t+\beta)]\, V_0\cos\omega t. \tag{54}$$

The first term represents the usual mass reactance and the second term a modification due to the fluctuating density.

Clearly, the time average over a cycle would show no change in the mass reactance, and it would be necessary to consider the higher harmonic terms before nett changes become evident. The analysis would follow similar lines to that of Zorumski where all the harmonic terms are taken into account. Rather than pursue such an analysis, it is considered more appropriate here to continue along the lines proposed by Westervelt [35] and to quantify more precisely the transition regions and the possible dependences.

It was established in section 2 that the mass reactance of the orifice consists of three components: the core mass and the two equal attached masses, one on either side of core mass. As a result of the measurements of Ingard and Labate [11] and the known steady flow behaviour of orifices, the flow on the upstream side (positive or driving pressure side) can be taken to be potential. Correspondingly, it may be assumed that the attached mass is undisturbed since it, itself, results from the potential flow field. However, on the downstream side the flow beyond the *vena contracta* is known to be turbulent, at least for Reynolds numbers in excess of 10 to 30 (see references [25] and [37]). Therefore, since Reynolds numbers in excess of approximately 10 mark the onset of turbulent flow (on the downstream side), it would be expected that the non-linear behaviour of the mass reactance would also start to become evident beyond this value. Further, at least on intuitive grounds, it would appear reasonable to assume that the location of the *vena contracta* also is significant in determining the extent of the non-linearity since presumably up to the *vena contracta* the flow is laminar and becomes turbulent beyond it. Correspondingly, in the laminar flow region beyond the hole and up to the *vena contracta* the mass reactance due to attached mass is presumed to exist and beyond this point is destroyed. Unfortunately, very little information on the location of the *vena contracta* appears to exist in the open literature. The descriptions are qualitative and vague: e.g., in reference [31] it is noted that "As the diameter ratio increases" (author's note: i.e., that of the orifice to containing flow tube or zone area) "the radial component of velocity decreases, and, therefore, the degree of jet contraction also decreases. This is accompanied by a movement of the *vena contracta* towards that of the orifice." This means that, for a given orifice geometry, for increasing flow velocities the attached mass will be more and more disturbed as the velocity increases. Further, in the case of perforates, the lower the porosity the sooner this effect will occur for a given hole velocity. Finally, from reference [37] it would appear that for Reynolds numbers in excess of 2000 fully developed turbulent flow exists beyond the orifice and the attached mass can be assumed to be completely destroyed. For the orifices, therefore, between Reynolds numbers of say 30 and 2000, the non-linearity in the mass reactance would be expected to depend on the inverse of the Reynolds number and the porosity. For thick orifices it might be expected that an additional factor based on the length to diameter ratio would be important because of the viscous damping of the radial velocity component. If, therefore, the change in mass reactance due to non-linear effects is written as Δm its functional dependence may be written in the form

$$\Delta m = \{\text{attached mass for one side}\}\, f\left(\frac{uD}{\nu}, \left(\frac{l}{D}\right), P\right), \qquad (55)$$

where f represents some function of the parameters indicated and the other symbols have their usual meanings. The first term in brackets is the attached mass for one side of the orifice corrected for interaction effects.

4.3. COMPARISON OF MEASURED AND THEORETICAL RESULTS

4.3.1. *Resistance*

Within section 4.2.2. an expression was presented—equation (52)—for the specific acoustic resistance of a perforate. In particular, the non-linear resistance was found to be dependent on the porosity, the discharge coefficient, and the acoustic particle velocity through the holes in the perforate. Correspondingly, for a given perforate, at least according to the model presented here, the results for the measured non-linear resistance at the various cavity depths (frequencies) should collapse on to a common graph. The results for each of the six perforates is therefore presented in this manner in Figures 31 to 36, Figures 31 to 33 being for the 22% perforates and Figures 34 to 36 for the 7½% perforates. It is evident from these figures that the non-linear resistance is apparently insensitive to frequency (a conclusion reached by Sivian [10]) and further it exhibits a linear relationship with velocity. The two other parameters determining the non-linear resistance are the porosity and the discharge coefficient. Since the discharge coefficient is also dependent on the porosity it is not practical, nor is it necessary, to investigate separately the dependence on these two parameters. Because of the uncertainty of both the effects of the coupling between orifices and of the finite orifice thickness on discharge coefficient, the latter, for a given perforate, changing with hole size (since the plate thickness is constant), the most reasonable method of interpreting the results would appear to be to establish the discharge coefficients and then compare these with the results for single, ideal orifices and also with values obtained from measurement of steady flows through the particular perforates. The data for this latter comparison was provided by Rolls-Royce Limited (Hucknall). This data is presented graphically in Figures 37(a) and 37(b), the former being for the 7½% perforates, the latter for the 22% perforates.

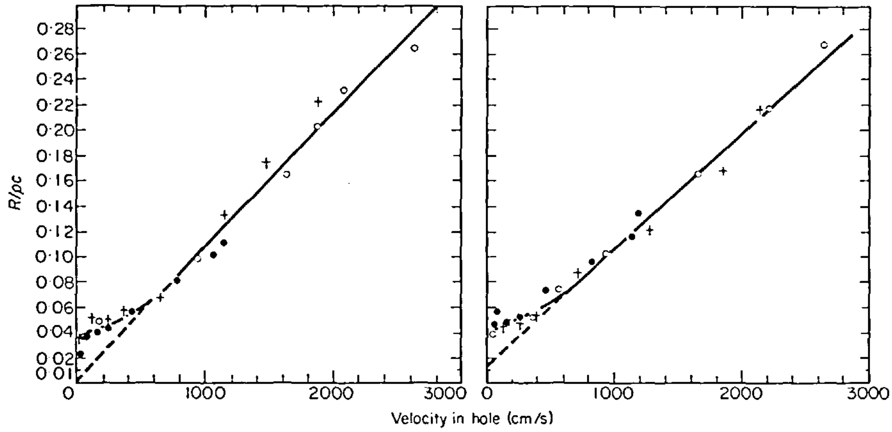

Figure 31. RT 335. Figure 32. RT 153/A.

Figures 31 and 32. Measured resistive impedance for two perforates. ○, $\lambda/4 = 3$ in; +, $\lambda/4 = 2$ in; ●, $\lambda/4 = 1$ in.

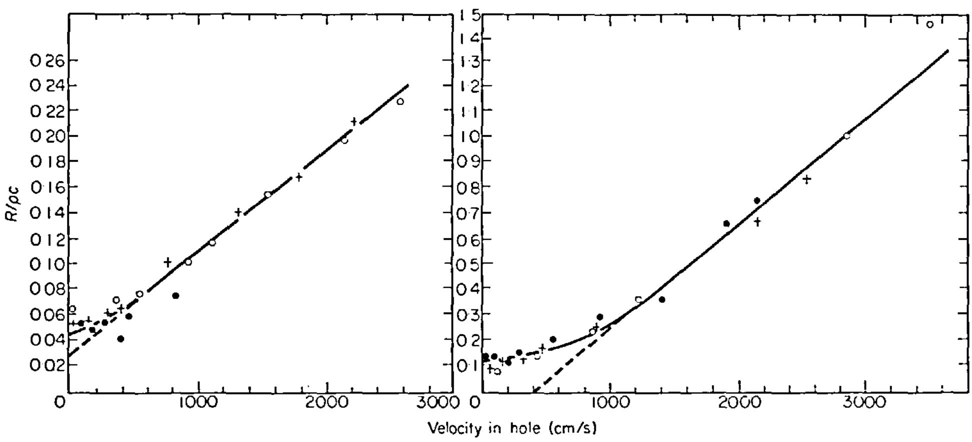

Figure 33. RT 233. Figure 34. RT 335/00.

Figures 33 and 34. Measured resistive impedance for two perforates. ○, $\lambda/4 = 3$ in; +, $\lambda/4 = 2$ in; ●, $\lambda/4 = 1$ in.

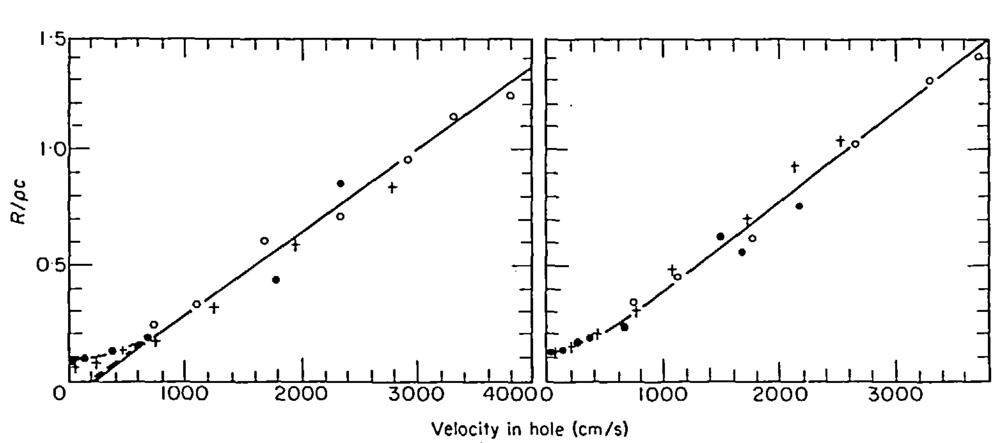

Figure 35. RT 153A/00. Figure 36. RT 233/00.

Figures 35 and 36. Measured resistive impedance for two perforates. ○, $\lambda/4 = 3$ in; +, $\lambda/4 = 2$ in; ●, $\lambda/4 = 1$ in.

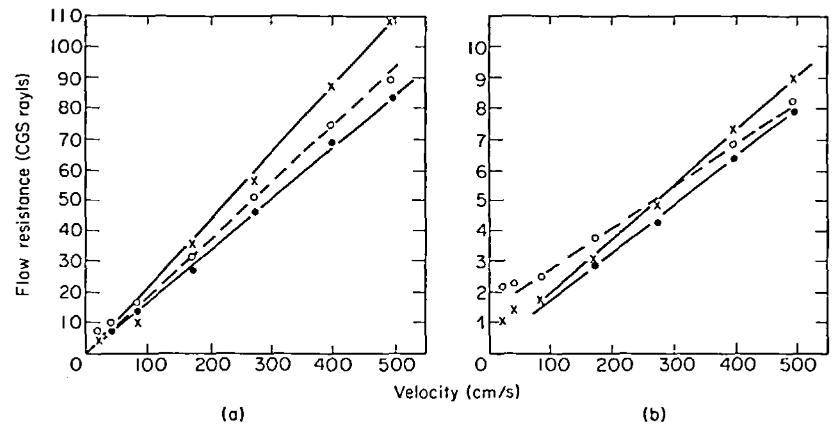

Figure 37. Plot of d.c. flow resistance *versus* velocity for (a) 7½% perforates; (b) 22% perforates (from Rolls-Royce data). ×, 0·101 in diameter, 0·22 in thick; ●, 0·050 in diameter, 0·22 in thick; ○, 0·026 in diameter, 0·018 in thick.

The derived discharge coefficients for the various perforates are presented in Tables 7 and 8. From the work of Johansen [37] it is known that the discharge coefficient for a single sharp-edged orifice is dependent both on the Reynolds number of the flow through the orifice and on the ratio of the orifice diameter to the pipe diameter in which it is mounted. In the case of perforates the latter quantity is interpreted as the orifice to zone area diameter: i.e., it is directly related to the porosity. For the 22% perforate this ratio is 0·46 and for the 7½% perforate 0·27. Johansen found for Reynolds numbers in excess of 10 that the larger the value of this ratio the larger was the discharge coefficient. The conclusion is borne out by the results in Tables 7 and 8, the 22% perforates having the larger value of discharge coefficient. Further, in general, the larger the value of this ratio the larger is the value of the Reynolds number before the discharge coefficient reaches its asymptotic value. For convenience the data of the appropriate figure from Johansen's work is presented in Figure 38. From the known acoustic particle velocity in the orifices the range of Reynolds numbers and, consequently, the range of discharge coefficients for each perforate, may be found (it being assumed, of course, that the behaviour is that of an ideal sharp edged orifice). The estimated ranges of values of the discharge coefficients are presented in Tables 7 and 8;

TABLE 7

Discharge coefficients, C_D, derived from acoustic and steady flow measurements for the 22% porosity perforates, together with value for a single ideal orifice of equivalent "porosity"

	Perforate code		
	RT 335 (110)†	RT 153 (50)†	RT 233 (26)†
Discharge coefficient from acoustic measurements (Figures 31–33)	0·845	0·894	0·955
Discharge coefficient from steady flow measurement (Rolls-Royce data: Figure 37(b))	0·802	0·854	0·925
Discharge coefficient for an ideal orifice (after Johansen [37])	0·72–0·65	0·72–0·65	0·72–0·65

† The quantity in parentheses is the hole diameter in thousandths of an inch.

TABLE 8

Discharge coefficients, C_D, derived from acoustic and steady flow measurements for the $7\frac{1}{2}$% porosity perforates, together with value for a single ideal orifice of equivalent "porosity"

	Perforate code		
	RT 335/00	RT 153A/00	RT 233/00
Discharge coefficient from acoustic measurements (Figures 34–36)	0·763	0·797	0·77
Discharge coefficient from steady flow measurement (Rolls-Royce data: Figure 37(a))	{0·705 0·745	0·804	0·764
Discharge coefficient for an ideal orifice (after Johansen [37])	0·69–0·62	0·69–0·62	0·69–0·62

only a single range has been given to cover all the holes although the lower limit will not be identical for all the hole sizes. Quite clearly, and perhaps not surprisingly, the discharge coefficient for the single ideal orifice is in both cases different from the value deduced from the acoustic measurements. Further, they are also considerably different from the values deduced from the steady flow measurements on the perforates. This clearly shows that either or both finite orifice thickness and interaction effects are important.

The discharge coefficient for the steady flow through the perforates is calculated from a

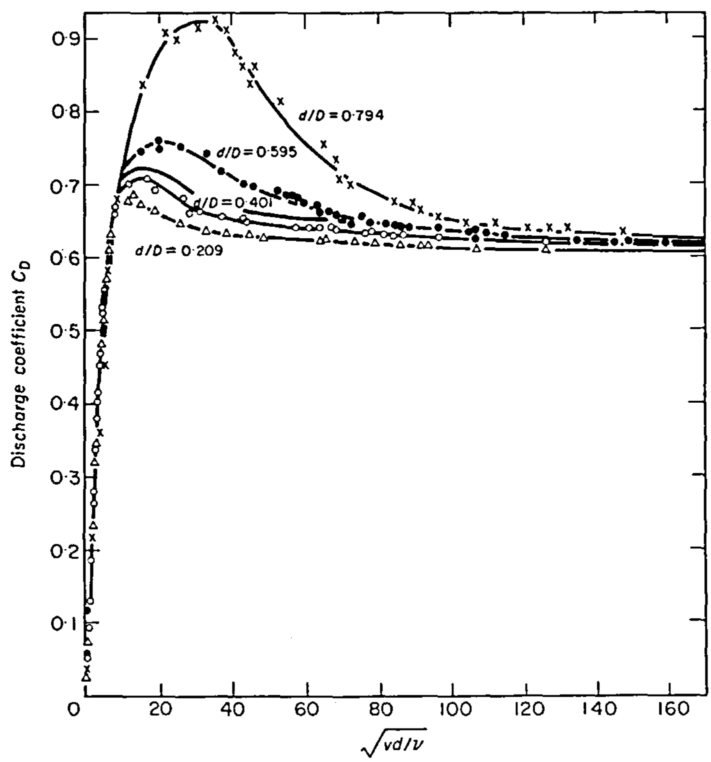

Figure 38. Discharge coefficients as a function of square root of Reynolds number, after Johansen [37].

suitable form of equation (49) used in conjunction with the results presented in Figures 37(a) and 37(b). The derived values, presented in Tables 7 and 8, when compared with the discharge coefficient derived from the acoustic measurements, show good agreement. For the 22% perforate the steady flow discharge coefficient is generally some 5% lower than the acoustically derived one. In the case of the $7\frac{1}{2}$% perforate the agreement is generally better, with the exception of the results for RT 335/00. In this instance the d.c. flow measurement could be interpreted as having one of two slopes for the resistance velocity characteristic; the discharge coefficient appropriate to both values has been quoted (note also that the hole diameter for the d.c. measurements is slightly less than for the a.c. case). The better agreement in the case of the lower porosity perforates is perhaps not surprising. The difficulties and possible errors involved in the determination of the pressure drop across the sample in the case of the d.c. measurements are more apparent the higher the sample's porosity.

Interaction effects between the flows through the orifices and/or finite thickness effects of the orifice are apparently very important. As an example of the importance of the effect, reference to Figures 37(a) and 37(b) shows the considerable differences in the slopes for constant porosity and plate thickness (this aspect will be referred to again later in the text).

From the results presented in Tables 7 and 8 it appears that the theoretical model derived in section 4.2.2. (equation (52)) for the non-linear resistance is a good approximation and self consistent, at least for the case of perforates. It is of considerable importance, therefore, to establish whether or not the model is valid for the case of a single ideal orifice and also, of course, a single non-ideal orifice. In the former case the acoustically derived discharge coefficient should correlate with the known values of discharge coefficient for steady flows (see, e.g., reference [37]), whereas in the latter case, if finite hole thickness effects are important, a significant difference would be expected.

In the work of Ingard and Ising [12] a graph of resistance *versus* velocity (Figure 21 of reference [12]) is presented for an orifice of 0·7 cm diameter where the plate thickness at the orifice is 0·01 cm; this, according to reference [31], may be taken as an "ideal" orifice. From the mean slope of the non-linear region and a suitable form of equation (52) (the results are presented as a function of velocity amplitude) the effective discharge coefficient may be deduced; the numerical value is 0·576. The hole to plate diameter ratio is 0·094; thus, according to Johansen [37], the discharge coefficient would be 0·608: i.e., the acoustically derived value is some 5% larger than the steady flow value. This result agrees with the findings for the perforates and can be taken as a further substantiation of the model.

Finally, for the case of the single non-ideal orifice, the results of Sivian [10] are relevant. In Figures 4a to 4d of reference [10] Sivian presents the measured resistance for a hole of approximately 0·09 inch diameter (cf. RT 335, 0·110 inch diameter) for orifice thicknesses of approximately 0·012 inch, 0·33 inch, 0·063 inch and 0·13 inch. It is interesting to note that for the case of the 0·033 inch thick orifice the non-linear region has a much greater slope than for the other cases; these other cases in comparison all have approximately the same slope. For comparison purposes it would be most appropriate, if possible, to choose a hole diameter and plate thickness close to the values for the perforates tested in this work. The hole diameter is fairly close to that of RT 335 whereas the plate thickness used in this study lies between two values used by Sivian: notably, the value used in the set of measurements was 0·022 inch whereas the closest values to this used by Sivian are 0·012 inch and 0·033 inch; further, as noted earlier, the behaviour for the case of the 0·033 inch thick plate is peculiar. Correspondingly, any interpretation as such is somewhat suspect since, at best, the results will be based on an interpolation of values for the discharge coefficient between the results of Sivian for the two plate thicknesses given in order to correlate with the plate thickness used in the tests contained herein. Further, of course, it is necessary to assume that the hole diameters are identical.

The values of the discharge coefficient calculated from Sivian's data for the plate thicknesses of 0·012 inch and 0·033 inch are, respectively, 0·790 and 0·512. When it is assumed that the $7\frac{1}{2}\%$ perforate RT 335 (0·110 inch hole diameter) should be approximately equivalent to these plates, the discharge coefficient derived from acoustic measurements is 0·763, whereas the discharge coefficient from steady flow measurements on the perforate is in the range of 0·705 to 0·745. Because of the spread in results, as determined by Sivian, for the different hole thicknesses, the conclusion drawn must be viewed with caution; however, for the thinnest plate used by Sivian the correlation between his results and the equivalent ones quoted herein are reasonable. Note that in this case (i.e., the thinnest of Sivian's plates) the plate thickness to hole diameter ratio is more near to the value for the sample quoted (RT 335) than for his next plate thickness.

Thus the comparison of the author's experimental results with predictions based on the semi-empirical theoretical model for non-linear resistance behaviour as developed herein (i.e., equation (52)) appears to provide satisfactory and consistent agreement. Further, as far as can be assessed, it also gives a satisfactory explanation of results when applied to measurements obtained by other authors; this is particularly encouraging in view of the differences in experimental techniques used and the possible sources of error which may arise in this type of measurement. Comparisons with other data would be highly desirable, of course. However, an extensive search failed to reveal any other data in the literature which were sufficiently accurate and well-defined for this purpose.

4.3.2. *Reactance*

The behaviour of the mass reactance is of lesser significance than the resistance, at least in the engineering application of perforates at high intensities, because of the small influence it normally has on the total impedance of any practical perforate-based absorber system (in this context thin perforates are implied). However, it is of general interest to establish the basic trends in the behaviour of this quantity since very little detailed work has been reported in the case of perforates. The analysis is based on the qualitative descriptions presented in section 4.2.3. The measured results are those for the two sets of perforates as described in section 3 (see, e.g., Figure 18). Whilst considerable scatter is evident in the results, the usual noted trend of the mass reactance of decreasing to approximately one-half its nominal value can be seen to occur; the reasons for this and the expected final values are well noted in the literature (see, for example, reference [35]). This has also been discussed at some length in section 4.2.3 and will not be pursued further here. Those characteristics of the behaviour which are of particular interest are the point of onset of the non-linear behaviour, and the point at which the lower limit of the mass reactance is reached. Previously, in section 4.2.3, possible criteria in terms of Reynolds number of the flow (based on hole diameter) were noted for the onset of these phenomena. In particular, a Reynolds number of approximately 30 was defined for the onset of the non-linear behaviour and a Reynolds number of 2000 was stated as defining the point at which the mass reactance would reach its lower limiting value. It is not proposed to examine in any detail the criterion for the onset of the non-linear behaviour both because the scatter in the data make detailed examination difficult and in any case the discussion in relation to the point where the mass reactance reaches its lower limiting value can probably be extended to that particular case.

On the Reynolds number criterion the mass reactance would reach its asymptotic lower limit at different velocities for the different perforates. Reference to the results, however, clearly shows that this is not the case in practice. In particular, reference to the 22 % perforate results (see, e.g., Figure 18) shows that for the three perforates in this group the asymptotic lower limit of the mass reactance is reached, in all cases, at an in-hole velocity in the region of 1200–1400 cm/s. On the Reynolds number based criterion the limiting values would be

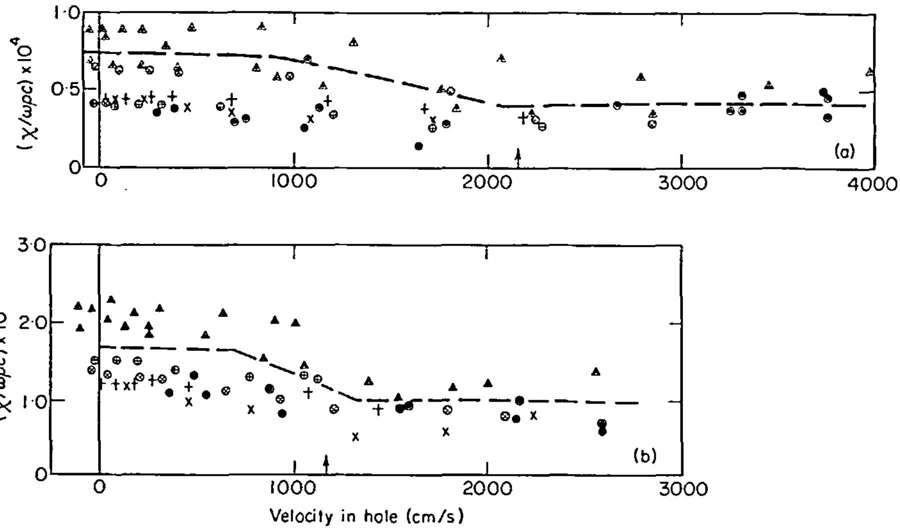

Figure 39. Measured mass reactance for the various samples plotted against in-hole velocity scaled on plate thickness to hole diameter ratio and square root of porosity. •, 3 in cavity; ×, 2 in cavity; +, 1 in cavity; O, RT 153A; plain, RT 233; △, RT 335. (a) 7½% perforate, (b) 22% perforate.

expected at velocities of 1250, 2400 and 4000 cm/s for the 0·110 (RT 335), 0·50 (RT 153A) and 0·026 (RT 233) inch hole diameter perforates, respectively. The only case showing any agreement is that of the large hole perforate, RT 335. However, it was noted in section 4.3.2 that the ratio of the plate thickness to the hole diameter would be expected to have some significant influence. Indeed, taking the results for a given porosity and modifying the velocity dependence, according to the thickness to diameter ratio, collapses the point at which the mass reactance reaches its minimum value to a common velocity. More specifically, the velocity at which the mass reactance reaches its asymptotic minimum is dependent on both the Reynolds number and the plate thickness to hole diameter ratio. When this factor is taken into account, then, for a given Reynolds number, the velocities for the three perforates RT 335, RT 153A and RT 233 are in the ratios 2·7:2·68:2·62: i.e., they are approximately the same as noted from the results. Accordingly, all the data for the 22% perforates is presented on a common curve in Figure 39. Finally, it should be noted that in the case of the 7½% porosity perforates the mass reactance attains its asymptotic lower value for in-hole velocities in the range 1800–2200 cm/s. This is considerably higher than that found for the case of the 22% perforates, but is perhaps not surprising. In the case of the lower porosity perforate, for a given flow velocity through the hole, the radial component of the velocity will be higher. This latter effect will result in an extension of the potential flow region (i.e., the flow can be considered to be forming a flow nozzle); correspondingly, when compared with the behaviour of a higher porosity perforate (other factors remaining constant) less disruption of the attached mass would be expected, or, in other words, the disruption of the attached mass would be expected to occur at a higher flow velocity through the hole. The radial component of the velocity will be proportional to the ratio of the hole diameter to the flow pipe (or zone) diameter (see section 4.2.3), and therefore proportional to the square root of the porosity. Thus, for the 7½% porosity perforates, in comparison with the 22% perforates, it is anticipated (other factors remaining constant) that the asymptotic lower limit of the mass reactance is reached at a velocity some $\sqrt{3}$ higher than in the latter case. The results for the 7½% perforates have been plotted on the same graph as those for the 22% in Figure 39. The velocity has been scaled according to the noted procedure based on porosity.

It is evident that, within the limits set by the scatter, the suggested dependence of the point at which the mass reactance reaches its asymptotic limit shows good agreement with the methods of scaling: that is, that velocity at which the limit is reached can be written in the form

$$\text{const} = \frac{u_H D}{v}\left(\frac{l}{D}\right)\sqrt{P},$$

where l is the plate thickness, D is the hole diameter, v is the kinematic viscosity, P the porosity and u_H the root mean square velocity in the hole.

In this particular instance the constant is about 220.

Within the limited range of the present results, and within the limitations set by the scatter in the data, it would appear that a method of predicting the point at which the mass reactance reaches its apparent asymptotic minimum value has been established. Whilst the evidence is by no means conclusive it does indicate what parameters are important and should be taken into consideration if a detailed study of the behaviour of the mass reactance of perforates is to be made.

4.4. GENERAL DISCUSSION

The major objective of the study has been to quantify in some detail the behaviour of the non-linear resistance of perforates in terms of well-known, or at least easily determined, parameters of the system. In particular, it has been demonstrated that the discharge coefficient is an important parameter. Further, the discharge coefficient must, in the case of perforates, be determined for each particular geometry: i.e., it is not sufficient or adequate to use a value based on an ideal or sharp-edged orifice. Of major importance in this study is the result that the acoustic resistance can be directly related to the steady or d.c. flow resistance through a constant factor, $8/3\pi$, based on velocity amplitude. Further, therefore, with a knowledge of the nominal value of the resistance the resistance/velocity curve for the perforate can be generated. It also follows that, for any given absorber configuration, the resistance as a function of some suitably defined pressure may be derived. It is also noteworthy that the particular model presented also satisfactorily quantifies the non-linear resistance/velocity dependence of single, sharp-edged orifices. Further, there are also indications that the theory will prove adequate for single orifices of finite thickness.

The behaviour of the mass reactance, whilst of lesser significance from an engineering standpoint because of its small contribution to the impedance of the system, has been examined in a semi-qualitative manner. The important parameters determining the lower limiting value of the mass reactance are the Reynolds number, the plate thickness to hole diameter ratio and the square root of the porosity. The study has indicated those parameters which seem important and provides guidelines for studies specifically aimed at the understanding of the non-linear behaviour of the mass reactance of perforates.

As a direct consequence of the success of the theoretical model in predicting the non-linear behaviour, it is reasonable to assume that the physical phenomena responsible are closely related to those observed for steady flow through orifices since, on the assumption of a quasi-steady analysis, the basic parametric dependence of orifice plates has been successfully carried over to the acoustic situation. Correspondingly, therefore, the major characteristics of the flow field behaviour can be evaluated [37]. In terms of detailed behaviour, the discussion of the parametric dependence of the mass reactance would suggest that the *vena contracta* and its spatial location with respect to the orifice is of considerable significance; indeed confirmation of its existence or otherwise (by means of, for example, hot-wire techniques) would prove a valuable aid in predicting the quantitative behaviour of the mass reactance.

Finally, at least in the context of the work contained herein, the Strouhal number has been shown to be an irrelevant and unhelpful parameter.

5. SUMMARY OF MAIN RESULTS

5.1. INTRODUCTION

The primary objective of the programme of work, as set forth in section 1.3, was to provide a practical method of predicting the acoustic behaviour of perforated materials, particularly the resistive part of the impedance. As a natural consequence other initially undefined but equally important objectives arose. These also required some investigation and understanding if the primary objective was to be achieved. Further, through the association of the work with the development of aero-engine intake attenuator configurations, other studies arose which in themselves had fixed objectives, even though the outcome was only loosely and indirectly associated with the prime objective. Correspondingly, therefore, the assessment of the results within the overall study can be conveniently examined in three main categories: (i) the prime objective; (ii) those resulting directly from the prime objective and (iii) those more loosely associated with the prime objective.

In this section, therefore, and as a conclusion to this report of the work, a summary of the main results will be presented, in accordance with the categories noted above, but in reverse order so that the results in relation to the prime objective are presented finally.

5.2. THE TWO-MICROPHONE METHOD OF IMPEDANCE MEASUREMENT

This method of impedance measurement could, in a broad sense, be considered as part of the major aim of the study through the use of the pressure measurement in the cavity to determine the acoustic particle velocity at the perforates. However, viewed in terms of the major use of the technique, for accurate impedance measurement, it turned out to be only loosely associated with the primary objective, as the transmission line method of impedance measurement was found to be preferable as well as having been selected *a priori.*

In combination with the method of data interpretation the programme of measurement was sufficiently comprehensive and detailed to demonstrate that the principal difficulty with the two microphone method was simply that the phase measuring system was inadequate (in spite of the manufacturer's claims). It was neither valid nor reasonable in the light of the findings to pursue in this programme of work the potential difficulties associated with this technique of impedance measurement. The method has subsequently been further developed by Dean [5] and successfully used for *in situ* measurement of impedance in ducts carrying mean flows.

5.3. INSERTION LOSS MEASUREMENT

The results of this particular study were included not because of any significant achievements of the programme, but rather because of some of the interesting questions which arose as a result of it, in particular, those questions relating to the modelling of the wall impedance where the sample impedance is non-linear; this changing wall impedance results from the non-linear attenuation of the propagating pressure wave in the treated duct. Subject, of course, to the limited range of configurations tested, the insertion loss measurements carried out provide some indication of the errors resulting from certain well-defined assumptions, and further the data and results presented may provide useful information for a more comprehensive study.

5.4. STUDIES ASSOCIATED WITH THE PRIME PROGRAMME OBJECTIVE

The three significant aspects of the work which will be discussed in this section are as follows: (i) the coupling of the acoustic and mechanical impedances of the test sample which was presented in section 3.2; (ii) the calculation of the nominal acoustic impedance of perforates discussed in section 2; (iii) the calculation of the non-linear impedance discussed in detail in section 4. These three items are discussed separately below.

5.4.1. *The coupling of the acoustic and mechanical impedances of the test sample*

Two significant factors emerged from the results of this study: (a) the mechanical impedance of the perforate could be disregarded; (b) the honeycomb in the backing cavity has a finite effect on the resistive impedance. The former effect (a) is limited in its applicability because the mechanical impedance exhibited by a perforate will, in general, be a function of its panel size, thickness and edge fixing conditions. However, the latter aspect of the work (b), the increase in resistance due to the presence of the honeycomb, is a new factor which has not been reported previously, even though its existence, based on well-known physical arguments, would be anticipated. As a direct consequence of its existence, and as demonstrated, modification of the nominal and non-linear regions of the resistive impedance of the perforate occur. Correspondingly, if the factor is not taken into account, interpretation of the behaviour of the impedance of perforates having honeycomb in the backing cavity could lead to erroneous conclusions. It was demonstrated, by using simple theoretical models, that both the contribution to the resistance and the shift on the abscissa of the non-linear region due to the honeycomb could, in general, be satisfactorily explained. The work is therefore of relevance to the typical in-engine configuration where the honeycomb behind the perforate is the rule, rather than the exception.

5.4.2. *The calculation of the nominal impedance of perforates*

As a direct consequence of this study significant and original improvements have been made in the theoretical formulae for the estimation of the nominal impedance of perforates. These improvements resulted from a detailed survey of the theoretical and experimental evidence available on single orifices, which was extended to perforates.

For single orifices it was concluded that the usual Rayleigh correction factor (viz., $8r_0/3\pi$, per end) for the effective increase in hole length per end was adequate when estimating the attached mass. In the case of resistance, an additional resistance term due to an effective extension of the orifice should be included, the effective length of this extension being, in fact, the same as that for the attached mass per end. However, for these extensions a value for the viscosity should be chosen consistent with that for the non-thermally conducting situation. For the case of perforates it was concluded that, as compared with the single orifice and for the situation where the fields are potential, the end correction terms should be modified for interaction effects between holes; the modifying function is of the form derived by Fok. Because particular and distinct conclusions were drawn in respect of the resistive and reactive impedance terms for perforates these are conveniently discussed separately.

(i) *Resistance.* In this case the comparison of calculated nominal impedances based on the Fok interaction function and the experimental results gave poor agreement.

However, the application of the Fok function is valid only when the flow in the hole is potential. Further, in this instance experimental results indicated that the nominal resistance value could not have been obtained for the perforates at the typical lower pressure levels at which the measurements were made. Correspondingly, no meaningful conclusion could be reached as to the validity of its application to resistance. Indeed, on the basis of the evidence for the results on mass reactance, indications are that if the resistance had been measured at sufficiently low sound pressure levels the correction would apply.

(ii) *Reactance.* Comparison of calculated nominal reactance values and measured results was found to give good agreement when the Fok interaction correction was applied. The agreement was found to be particularly good for the cases where the end correction controlled the mass reactance. In this case neglecting the interaction function could lead to errors of some 100%. The error is, of course, dependent on how much the attached mass contributes to the total effective mass of the orifice. The lower sensitivity of the reactance to velocity through the orifice would appear to be reasonable, as was explained in the qualitative discussion presented in section 4.

The results can be formally summarized as follows. The general equation for the nominal normal specific acoustic impedance of a perforate of porosity P when radiation resistance is negligible, may be written as (see equation (18))

$$z = \frac{1}{P}\,[l_1(j\omega\rho)/\square(k_s r_0) + 16 r_0(j\omega\rho)/3\pi\square\,(k_s' r_0)\,\psi'(\xi)],$$

where the symbols have their usual meanings. This equation is the best obtainable at present, on both theoretical and experimental grounds. When assessed on the basis of the experimental work described here, it represents a considerable improvement over previously available formulations.

5.4.3. *The prediction of the non-linear impedance of perforates*

The primary objectives of the work, as stated in section 1.3, were as follows: (i) to provide a practical method of prediction of the liner behaviour using readily available engineering data; (ii) to provide physical insight into the mechanism of non-linear behaviour *via* a detailed, controlled, precision measurement programme. Naturally, as a direct consequence of (i) it was necessary to provide a theoretical model for non-linear behaviour. As is evident from section 4 these objectives have been achieved; however, it is pertinent to examine these in some detail. The discussions in relation to reactance and resistance, respectively, are dealt with separately below.

(i) *Reactance.* The significant improvement in the understanding of the behaviour of the mass reactance has resulted through the identification of those parameters which are important in defining the point at which the mass reactance takes on its lower limiting value as a result of the non-linear phenomena. The three important parameters identified are (a) the Reynolds number based on hole diameter, (b) the ratio of the plate thickness to hole diameter and (c) the square root of the porosity. From the range of experiments concluded herein, a combination of these factors was found which could be equated to a constant: namely,

$$220 \simeq \left(\frac{u_H D}{\nu}\right)\left(\frac{l}{D}\right)\sqrt{P},$$

where the symbols have their usual meanings. Through the related discussion in section 4 and on the assumption of a quasi-steady model this particular parametric dependence indicates that the location of the *vena contracta* in relation to the sample faces is of considerable importance. Correspondingly, if more detail of the spatial behaviour of the flow through the hole were known, quantitative estimations of the progressive loss in attached mass could be made.

(ii) *Resistance.* A convenient and practical method of prediction of the non-linear behaviour of perforates has been evolved.

Its originality stems from the development of the description of the behaviour of the fluid in the orifice through the fundamental momentum equation coupled with the interpretation of the behaviour as a quasi-steady phenomena; the consequent identification of the relevant parameter in the steady flow situation and the application of conservation of energy ultimately yield the desired relationships. Comparison of the theory with experimental results has demonstrated that not only does it consistently explain the results contained herein but adequately predicts the performance of an ideal, isolated orifice, and also, as far as can be established, that of isolated, non-ideal orifices.

It has been clearly demonstrated that, specifically, the discharge coefficient is the most important parameter and further, this parameter, as was to be expected from the work on steady flows, is sensitive to orifice thickness to hole diameter. It is also evident that in the case of perforates it is sensitive to porosity.

In practical terms, it is not necessary to determine the discharge coefficient, but rather the acoustic non-linear resistance can be derived directly from the steady flow resistance (or pressure drop) velocity characteristics simply by its multiplication by an appropriate constant. In terms of the root mean square particle velocity in the orifice (hole) u_{H}, the relationship between the specific acoustic resistance ratio, R, and the slope of the steady flow resistance velocity curve, M, is

$$R = \frac{1}{\rho c}(R_\mu + 1{\cdot}2Mu_{\mathrm{H}}),$$

where R_μ is the nominal specific acoustic resistance, other symbols having their usual meanings.

It is perhaps interesting to note that since the discharge coefficient is sensitive to hole shape (that is, for example, whether or not the holes are rounded) and because the characteristics of the non-linear region ultimately depend on the discharge coefficient, then presumably a limited amount of modification to the non-linear resistance behaviour can be engineered (e.g., the discharge coefficient of a sharp-edged orifice is lower than that of a venturi; correspondingly, the venturi would have a less steep resistance/velocity characteristic).

This simple empirical relationship between the acoustic non-linear behaviour and steady flow behaviour should extend up to the condition where the orifice becomes choked, a condition which is unlikely to be obtained in practice. Over this range the error in the slope of the resistance/velocity curve should be no more than 10% (since, as noted, the discharge coefficients may be in error by 5%).

Finally, the problem arises of the application of this formulation in practice where pressures rather than velocities are dealt with. The ratio of the total pressure to velocity at the surface of the absorbing configuration, must, by definition, be equal to the impedance. Further, for both the incoming and outgoing waves (incident and reflected), the ratio of pressure to velocity must be equal to the characteristic impedance of air. Correspondingly, through suitable manipulation of the equations, it can be shown that the total velocity at the surface of the absorber is equal to twice the incident pressure divided by the series sum of the specific impedance of the liner (velocity sensitive) plus the characteristic impedance of the air. Consequently by a suitable iteration procedure (given the velocity dependence of the perforate impedance) the velocity can be calculated for a given incident pressure and hence the impedance determined.

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REFERENCES

1. P. MUNGUR and G. M. L. GLADWELL 1969 *Journal of Sound and Vibration* **9**, 28–48. Acoustic wave propagation in a sheared fluid contained in a duct.
2. P. G. VAIDYA 1969 *Paper presented at the 78th Meeting of the Acoustical Society of America.* Influence of flow and high sound levels on the attenuation in a lined duct.
3. P. MUNGUR 1970 Personal communication.
4. B. J. TESTER 1972 *Ph.D. Thesis, University of Southampton.* Sound attenuation in lined ducts containing subsonic mean flow. (See also 1973 *Journal of Sound and Vibration* **28**, 151–203. The propagation and attenuation of sound in lined ducts containing uniform or "plug" flow.)
5. P. D. DEAN 1972 *Ph.D. Thesis, University of Southampton.* On the measurement of the local impedance of the walls of flow ducts and its use in predicting sound attenuation.
6. L. CREMER 1953 *Acustica* **3**, 249–263. Theorie der luftschalldampfung im rechteckkanal mit schluckender wand und das sich dabei ergebende hochste dampfungsmass.
7. B. J. TESTER 1970 Personal communication. (See also 1973 *Journal of Sound and Vibration* **27**, 477–513. The optimization of modal sound attenuation in ducts, in the absence of mean flow.)
8. T. H. MELLING 1968 *Lockheed-Georgia Company, Final Report, Vol. IV.* Aircraft exterior noise control.
9. T. H. MELLING and P. E. DOAK 1971 *Journal of Sound and Vibration* **14**, 23–35. Basic design considerations and theoretical analysis of double-reverberant chamber duct lining test facilities.
10. L. J. SIVIAN 1935 *Journal of the Acoustical Society of America* **7**, 94–101. Acoustic impedance of small orifices.
11. U. INGARD and S. LABATE 1950 *Journal of the Acoustical Society of America* **22**, 211–219. Acoustic circulation effects and the nonlinear impedance of orifices.
12. U. INGARD and H. ISING 1967 *Journal of the Acoustical Society of America* **42**, 6–17. Acoustic nonlinearity of an orifice.
13. W. C. SPERRY 1966 *Paper presented at the 72nd Meeting of the Acoustical Society of America.* Reduction of turbomachinery noise by absorption and choked flow.
14. C. ZWIKKER and C. W. KOSTEN 1949 *Sound Absorbing Materials.* New York: Elsevier Publishing Co.
15. T. H. MELLING 1973 *Journal of Sound and Vibration* **28**, 23–54. An impedance tube for precision measurement of acoustic impedance and insertion loss at high sound pressure levels.
16. T. H. MELLING 1972 *Ph.D. Thesis, University of Southampton.* The acoustic impedance of perforates at medium and high sound pressure levels.
17. A. LONDON 1950 *Journal of the Acoustical Society of America* **22**, 263–269. Determination of reverberant sound absorption coefficients from acoustic impedance measurements.
18. LORD RAYLEIGH 1940 *Theory of Sound.* Vol. II. London: Macmillan & Co. Ltd.
19. I. B. CRANDALL 1927 *Theory of Vibrating Systems and Sound.* New York: D. Van Nostrand & Co. Inc.
20. U. INGARD 1953 *Journal of the Acoustical Society of America* **25**, 1037–1062. On the theory and design of acoustic resonators.
21. D. A. BIES and O. B. WILSON 1957 *Journal of the Acoustical Society of America* **29**, 711–714. Acoustic impedance of a Helmholtz resonator at very high amplitude.
22. V. A. FOK 1941 *Doklady̆ Akademii nauk SSSR* **31**. (In Russian.) Alternatively, see S. N. RSCHEVKIN 1963 *A Course of Lectures on the Theory of Sound.* London: Pergamon Press.
23. S. N. RZHEVKIN 1951 *National Advisory Committee for Aeronautics, Technical Memorandum* 1273. Resonance sound absorber with yielding wall.
24. R. H. BOLT, S. LABATE and U. INGARD 1949 *Journal of the Acoustical Society of America* **21**, 94–97. The acoustic reactance of small circular orifices.
25. J. K. WOOD 1954 *Journal of the Acoustical Society of America* **26**, 492–494. The acoustic resistance of a pipe orifice to steady-state fluid flow.
26. A. W. BLACKMAN 1960 (November) *American Rocket Society Journal*, 1022–1028. Effect of nonlinear losses on the design of absorbers for combustion instabilities.
27. W. E. ZORUMSKI and T. I. PARROTT 1970 *NASA SP*-189. Nonlinear acoustic theory for thin porous sheets.
28. U. INGARD 1970 *Journal of the Acoustical Society of America* **48**, 32–33. Nonlinear distortion of sound transmitted through an orifice.
29. W. A. SIRIGNANO 1966 *Princeton University A.M.S. Report* 553-*F* 31–40. Non-linear dissipation in acoustic liners.

30. B. T. ZINN 1970 *Journal of Sound and Vibration* **13**, 347–356. A theoretical study of non-linear damping by Helmholtz resonators.
31. American Society of Mechanical Engineers *Research Report on Fluid Meters: Theory and Application.*
32. G. K. BATCHELOR 1967 *Introduction to Fluid Dynamics.* Cambridge University Press.
33. A. H. GIBSON 1954 *Hydraulics and Its Applications.* London: Constable and Company Limited.
34. S. N. RSCHEVKIN 1963 *A Course of Lectures on the Theory of Sound.* London: Pergamon Press.
35. P. J. WESTERVELT 1951 *Journal of the Acoustical Society of America* **23**, 347–348. Acoustical impedance in terms of energy functions.
36. V. L. STREETER 1961 *Handbook of Fluid Dynamics.* New York: McGraw-Hill.
37. F. C. JOHANSEN 1927 *Proceedings of the Royal Society*, Series A, 116. Flow through pipe orifices at low Reynolds numbers.